

Precision Bipolar Coil Current Driver

Design, Failure Analysis, Simulation and Bring-Up of a Discrete
Class-AB BJT Output Stage for Magnetic Bias Coils

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Abstract

This document records the complete design, analysis and bring-up of a bipolar linear current driver for the magnetic bias coils (X, Y, Z and MOT axes) of a cold-atom magneto-optical trap experiment. The driver is a voltage-controlled voltage source built from a precision high-voltage operational amplifier (OPA455) followed by a two-stage complementary emitter-follower output stage using five paralleled MJL21194/MJL21193G transistor pairs. The current loop is closed externally by an SRS SIM960 analog PID controller reading a Danisense DP50IP fluxgate transducer.

The board did not function on first power-up. Four independent faults were identified and cleared: a floating enable pin that held the operational amplifier’s output stage disabled; a net tie with no assigned footprint, leaving the signal and power grounds unbonded; a parasitic oscillation near 3 MHz caused by the absence of base stopper resistors on the driver transistors; and a bias-generator floor that prevented the quiescent trimmer from reaching any useful operating point. A SPICE model built from the board netlist, vendor device models and PCB-derived trace parasitics reproduced the measured oscillation to within four percent in frequency and three percent in amplitude, and correctly predicted the base stopper value required for stability across the full command range.

Following the four corrections the board operates as designed: no oscillation, trimmable quiescent current settling at 50 mA warm, and a linear, in-phase transfer characteristic from 10 Hz to 1 kHz. A full netlist design audit produced eighteen findings that define the requirements for the next board revision.

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1 Introduction and Requirements

1.1 Purpose

The apparatus generates stable, fast-switching, low-ripple current through four sets of magnetic bias coils in a cold-atom / magneto-optical trap experiment. Three axes (X, Y, Z) provide bias and shim fields; a fourth drives the MOT gradient coils. Each axis is commanded by an analog setpoint voltage and must deliver bidirectional current through zero without a polarity-switching discontinuity, because bias-cancellation coils routinely operate near zero amperes.

1.2 Load and performance targets

Parameter	Value	Source
Coil inductance	0.886 mH	derived from measured τ and R_{dc}
Coil DC resistance	0.656 Ω	measured
Coil time constant $\tau = L/R$	1.35 ms	measured
Coil electrical pole	118 Hz	$R/(2\pi L)$
Target current	± 10 A	experiment requirement
Transducer ceiling	12.5 A	DP50IP at 4 primary turns
Stored energy at 10 A	44 mJ	$\frac{1}{2}LI^2$
DC compliance at 10 A	6.56 V	$I \cdot R_{dc}$
Zero crossing	continuous, no dead-time	experiment requirement
Original response target	500 μ s at 1 A–2 A	thesis specification

Table 1: Load parameters and performance targets.

1.3 Fundamental speed limit

Current slew into an inductive load is set by available forcing voltage, not by amplifier bandwidth:

$$\Delta t = \frac{L \Delta i}{V_{\text{forcing}}} \quad (1)$$

Rail voltage	10 A step time	Largest step in 100 μ s
± 15 V	591 μ s	1.7 A
± 30 V	295 μ s	3.4 A
± 50 V	177 μ s	5.6 A
± 90 V	89 μ s	10 A

Table 2: Achievable step times against rail voltage. A sub-100 μ s full-scale step is not achievable on ± 15 V rails; the DC compliance requirement of 6.56 V is easily met, so the limitation is purely dynamic.

This distinction governs several later conclusions. The output stage is not the speed bottleneck; the rail voltage is. Raising loop bandwidth beyond a few kilohertz buys faster small-signal response but cannot accelerate large steps.

1.4 Instruments in the control loop

SRS SIM960 analog PID

Input range ± 10 V common-mode, ± 1 V differential; 100 kHz bandwidth; 1 μ s typical propagation delay; 8 nV/ $\sqrt{\text{Hz}}$ input-referred noise above 10 Hz; proportional gain 10^{-1} to 10^3 ; integral

gain 10^{-1} to 10^5 s^{-1} ; $\pm 10 \text{ V}$ output with user-settable clamps; conditional-integration anti-windup; 200 ppm/ $^{\circ}\text{C}$ stability.

Danisense DP50IP

Closed-loop fluxgate transducer, wired for four primary turns (1:250 ratio, 12.5 A nominal). Linearity error 10 ppm maximum; offset 100 ppm including earth field; offset drift ≤ 1 ppm per month, ≤ 2 ppm per year; response time $1 \mu\text{s}$ at $100 \text{ A } \mu\text{s}^{-1}$; bandwidth 50 kHz (-0.1 dB), 300 kHz (-1 dB), 1 MHz (-3 dB); supply range $\pm 14.25 \text{ V}$ to $\pm 15.75 \text{ V}$; maximum burden 100Ω . With a 100Ω burden the scale factor is 0.4 V A^{-1} , equivalently 2.5 A V^{-1} .

Setpoint source

National Instruments DAQ analog output.

Power supplies

Four Delta Elektronika ES 015-10 (0 V to 15 V, 0 A to 10 A), floating outputs isolated to 600 V DC, arranged as two series pairs to form main $\pm 15 \text{ V}$ (PGND) and operational-amplifier/sensor $\pm 15 \text{ V}$ (SGND).

1.5 Document scope

Sections 2 and 3 cover the legacy prototype and the reasons it was replaced. Sections 4 through 8 cover the present board: design, audit, debugging, simulation and the corrections that made it work. Section 9 records measured performance. Sections 10 through 13 cover architectural assessment, next-revision requirements, retracted conclusions and reference calculations.

2 Legacy Prototype and Its Retirement

The present board replaces an earlier four-axis prototype. That prototype is recorded here because several of its design decisions, failures and measured numbers directly motivated the Phase 4 architecture.

2.1 Architecture

The legacy unit was a high-power linear voltage-controlled current source. A series MOSFET operated in its linear region as a voltage-controlled variable resistor; an external analog PID controller (SRS LTS-series) closed the loop around a current-transducer measurement to hold current constant regardless of coil impedance or thermal drift. No switching stage was used, so there was no PWM noise.

Signal path:

30 V supply $\rightarrow$ high-side MOSFET drain $\rightarrow$ series IGBT (static on/off switch) $\rightarrow$ coil load $\rightarrow$ LEM current transducer $\rightarrow$ PGND

The transducer secondary (1:250 ratio, 4 mA per primary ampere) drove a $100\,\Omega$ burden resistor, producing $0.4\,\text{V A}^{-1}$ of feedback to the external PID. The PID output drove the MOSFET gate directly, unbuffered. This single decision determined the prototype's speed limit, as Section 2.7 shows.

2.2 Hardware configuration

Four physical units shared one metal enclosure: X-bias, Y-bias, Z-bias and MOT. Construction was through-hole breadboard, hand-wired. Signals used BNC connectors (Feedback, Measure, PWM, DISABLE); power used banana plugs. Two cooling fans were fitted, with the IGBT gate drivers mounted on the enclosure walls.

Per-axis inputs were: main power 5 V to 30 V; 12 V logic; 5 V logic; and $\pm 15\,\text{V}$ bipolar for the current transducers. Swapping the 5 V and 12 V lines, or reversing either polarity, destroys the IGBT driver. As built, the Z-bias and MOT axes shared a single $\pm 15\,\text{V}$ line for their transducers, so a second $\pm 15\,\text{V}$ supply must not be wired to those two axes.

2.3 Grounding architecture

On the X-bias and Z-bias axes only, the main power negative return was physically tied to the 12 V logic ground. On all axes, the external Measure ground was internally tied to the current transducer's internal ground. Because of these ties, the external supplies must have floating outputs; an earth-ground path combined with the internal ties creates a ground loop. This constraint carried forward into Phase 4 and is the reason the present board uses floating Delta Elektronika supplies and a deliberate single-point SGND/PGND bond.

2.4 Absolute maximum ratings

Transfer function: 1 A of output corresponded to 0.4 V of setpoint input, a gain of $2.5\,\text{A V}^{-1}$.

2.5 Standard operating procedure

1. Fit T-adaptor oscilloscope taps on the Measure and Setpoint lines.

Parameter	Limit
Maximum load current	12.5 A (hard limit of the internal transducer)
Maximum main power voltage	100 V
Logic rails	strictly 5 V / 12 V / ± 15 V

Table 3: Legacy prototype absolute maximum ratings. Over-voltage on any logic rail destroys the front-end integrated circuits and the IGBT drivers.

2. Energise 12 V, 5 V and ± 15 V first, then the external PID controller.
3. Clamp the PID output to 0 V to 10 V *before* energising main power. Known-good baseline: $P = 0.8$, $I = 2.6 \times 10^5$, $D = 0$, offset 4.5 V.
4. Fans on, energise main power, apply setpoint.
5. Shut down in order: PID controller off, main power off, all logic and peripheral power off. Never hot-plug the coil, power or logic while energised.

2.6 Protection circuitry

A flyback diode was fitted across the series combination of coil and current transducer. A metal-oxide varistor was placed in parallel with that diode to clamp transients beyond the diode’s reverse breakdown. A ground-clamping diode tied the transducer output to ground, preventing negative excursions that would damage the transducer.

2.7 Measured performance

Rise time 26 μ s, fall time 30 μ s. This matches theory directly: the MOSFET gate charge $Q_g \approx 610$ nC divided by the PID’s maximum output current of 20 mA gives a theoretical minimum of 30.5 μ s.

$$t_{\min} = \frac{Q_g}{I_{\text{drive}}} = \frac{610 \text{ nC}}{20 \text{ mA}} = 30.5 \mu\text{s} \quad (2)$$

The agreement confirms that the unbuffered gate drive, not the coil or the control loop, was the bottleneck. The thesis target had been 500 μ s at 1 A to 2 A, which the prototype exceeded by more than an order of magnitude.

Response-time re-verification during recovery gave 400 mV $\rightarrow$ 1 A and 800 mV $\rightarrow$ 2 A, consistent with the 2.5 A V⁻¹ scale, with approximately 30 μ s rise time on all axes even with a coil connected. A 5.5 A step-response replication at 3 kHz and 5 kHz showed visible phase lag and oscillation approaching the gate RC limit, with roughly 10 % overshoot on transitions, consistent with the original thesis.

2.8 Failures and as-built modifications

X-bias IGBT gate driver failure

The EVAL-ADUM4221-1EBZ failed, leaving the IGBT stuck open. The driver drew 0.47 A and behaved as a short circuit. *Fix applied:* a bypass jumper was soldered from the MOSFET source directly to the IGBT emitter, removing the IGBT from the current path. X-bias then operated normally without its static switch.

Y-axis heating incident

Driving a 3.3 k Ω coil stand-in with a 770 ms, 0 mV to 272 mV square wave (approximately 20 mA) produced smoke after 20 s. The root cause was never established. Connections were re-checked and the Y axis was subsequently confirmed working (16 ms response at 1 V, $P = 0.7$).

Baseline good-axis behaviour

Y, Z and MOT all worked at a 1 kHz setpoint, 405 mV high and 0 V low, 50 % duty.

2.9 Legacy bill of materials

Component	Part number	Role
Linear MOSFET	IXTK60N50L2	500 V/60 A, main control element
IGBT	AUIRGPS4070D0	400 V/70 A, static series switch
IGBT driver	EVAL-ADUM4221-1EBZ	isolated half-bridge gate driver
Current transducer	LEM ITN 12-P ULTRASLAB	12.5 A, 1:250, closed loop
Burden resistor	100 Ω precision	0.4 V A ⁻¹ conversion
Reference resistors	3.3 k Ω	reference voltage dividers
TVS diode	5KP58A	5000 W transient suppressor
Varistor	V150LA20AP	150 V RMS MOV

Table 4: Legacy prototype bill of materials.

2.10 Redesign directions explored and abandoned

Before the Phase 4 architecture was chosen, several routes were evaluated and rejected. They are recorded so they are not re-explored.

- **Gate buffer.** A BUF634A or LT1210 between the PID output and the MOSFET gate would remove the 30 μ s charge-time bottleneck, theoretically reaching below 2.4 μ s with a 200 mA-class buffer.
- **Buffer requires a low-side MOSFET topology.** A high-side MOSFET has a floating source, which a ground-referenced buffer cannot drive. Adding the buffer therefore forced a topology change, not a component swap.
- **IGBT necessity.** No experimental procedure ever toggled the IGBT as a switch; the PWM and DISABLE pins were hardwired via jumpers. Removing it would save approximately 0.6 W A⁻¹ of conduction loss.
- **H-bridge candidates.** Pololu G2 24v21 and Infineon IM06B20AC1 intelligent power module (five samples ordered). Not pursued: an H-bridge reintroduces switching noise and a polarity-reversal discontinuity at zero current, which is unacceptable for bias-cancellation coils.
- **LT1210 buffer reference design,** if this path is ever revisited: non-inverting gain of one, $R_F = 750 \Omega$ (1 %, 0.25 W metal film), $C_{\text{COMP}} = 10 \text{ nF}$ from OUT to COMP, $R_{\text{PD}} = 10 \text{ k}\Omega$ gate pull-down, 100 nF plus 4.7 μ F on both rails, 10 Ω gate resistor.
- **Open question at the time.** Diode D1 in the original schematic did not match a standard flyback topology. This was raised with the thesis author; no reply is recorded.

3 Phase 4 Design: Discrete Class-AB BJT Amplifier

3.1 Topology and design intent

The Phase 4 board is a *voltage*-controlled voltage source, not a current source. A precision high-voltage operational amplifier provides gain and accuracy; a two-stage complementary emitter-follower discrete BJT output stage provides current capacity. The current loop is closed *externally* by the SIM960 PID reading the DP50IP transducer.

This is a deliberate architectural choice and it explains several features of the board that would otherwise look like errors. In particular, the operational amplifier's inverting input is tied to its own output, making it a unity-gain voltage buffer with the output stage outside its local loop. The consequence is a fixed offset of two base-emitter drops between the command and the load, which the external PID absorbs. Section 4.1 discusses the cost of this choice.

Signal path:

COMMAND → OPA455 non-inverting input → HV_DRIVE_NODE → Q12 V_{BE} -multiplier bias network → Q13/Q14 driver pair → five paralleled complementary output pairs → COIL_DRIVE → coil → DP50IP transducer → PGND

3.2 Node structure as verified against the netlist

The following was extracted from BJT_PUSH_PULL.net (KiCad Eeschema 10.0.4, 80 components, 46 nets). It differs from the structure assumed at the start of debugging, and that difference drove most of the audit findings.

/HV_DRIVE_NODE = U2.6 (OUT), U2.2 (−IN), Q13.1 (B), Q12.2 (C), R2.1, C22.2, D5.K, D6.K

There is no separate NPN_PREDRIVE net. The operational amplifier output, its own inverting input, Q13's base and Q12's collector are all one node. Two consequences follow immediately: the 1 μ F capacitor C22 hangs directly on the amplifier output with nothing in series, and the driver base capacitance is presented to the amplifier unisolated.

The V_{BE} multiplier divider chain:

```
HV_DRIVE_NODE --R2(10k)-- [X] --R23(2.2k)-- [Y] --RV2(5k trim)-- PNP_PREDRIVE
      |
      R26(10 ohm)    <- base stopper only
      |
      Q12.3 (B)
```

R26 is a base stopper for Q12. It is *not* an isolation resistor between the amplifier output and the predrive node. An early analysis assumed the latter, and the distinction matters: with R26 correctly understood, nothing isolates the amplifier output from C22.

Remaining nets:

/PNP_PREDRIVE = Q12.1 (E), Q14.1 (B), C22.1, D5.A, D7.A, R24.1, RV2.2, RV2.3
 /NPN_DRIVE_BUS = Q13.3 (E), R1.1, D3.K, D6.A, R6/R9/R13/R17/R21
 /PNP_DRIVE_BUS = Q14.3 (E), R1.2, D4.A, D7.K, R3/R10/R14/R18/R22
 /COIL_DRIVE = J8, D1.A, D2.K, D3.A, D4.K, all ten 0.22 Ω ballasts

3.3 Key stages

U2 — OPA455IDDAR

High-voltage precision operational amplifier, rated $\pm 6\text{ V}$ to $\pm 75\text{ V}$, 45 mA output, unity-gain stable. The PowerPAD is tied to V_- , correct per datasheet. Rated capacitive load drive is 200 pF.

Q12 — BD139 V_{BE} multiplier

Sets the class-AB quiescent bias. R2 = 10 k Ω top; R23 (2.2 k Ω) plus RV2 (5 k Ω , 25-turn Bourns 3296W) bottom. Mounted in the heatsink row between Q10 and Q5 for thermal tracking.

Q13, Q14 — MJE15032 / MJE15033, TO-220

First-stage complementary driver pair.

Q2–Q11 — $5\times$ MJL21194 NPN, $5\times$ MJL21193G PNP, TO-264

Paralleled output stage. 2.2 Ω base resistors (2512 SMD), 0.22 Ω emitter ballasts (TO-220-2 power resistors).

U1 — Danisense DP50IP

Fluxgate current transducer wired for four primary turns, giving a 1:250 ratio and 12.5 A nominal. R25 = 100 Ω burden to SGND, output to BNC J13.

D1, D2 — TO-220-2

Specified as MUR1560G-class, 600 V 15 A. Clamp COIL_DRIVE to the $\pm 30\text{ V}$ rails, backed by two 1000 μF capacitors per rail.

R1 — 100 Ω

Driver-stage bleed between the two drive buses, approximately 12 mA. Its purpose is to keep both drivers conducting a defined current so neither fully cuts off at crossover.

Decoupling

Two 1000 μF plus five 100 nF on each main rail, symmetric between rails; 4.7 μF plus 100 nF on each amplifier rail.

3.4 Reverse V_{EBO} protection network

The MJL21193/94, MJE15032/33 and BD139 all specify $V_{EBO} = 5\text{ V}$ despite V_{CEO} and V_{CBO} ratings of 250 V to 400 V. This asymmetry is the reason for a dedicated protection network.

During power cycling, the small amplifier-rail capacitors (4.7 μF) collapse far faster than the 1000 μF main-rail capacitors. The predrive network can therefore be driven to a potential that reverse-biases base-emitter junctions past 5 V. Reverse base-emitter breakdown is not usually immediately destructive, but it cumulatively degrades h_{FE} , and in a five-way paralleled bank an h_{FE} shift means a current-sharing shift. Five diodes clamp each vulnerable junction to one forward drop.

Orientation was confirmed directly from the netlist, which supersedes all earlier screenshot-based analysis.

Diode	Anode net	Cathode net	Required anode	Verdict
D3 (SS34)	COIL_DRIVE	NPN_DRIVE_BUS	COIL_DRIVE	correct
D4 (SS34)	PNP_DRIVE_BUS	COIL_DRIVE	PNP_DRIVE_BUS	correct
D5 (1N4148W)	PNP_PREDRIVE	HV_DRIVE_NODE	PNP_PREDRIVE	correct
D6 (1N4148W)	NPN_DRIVE_BUS	HV_DRIVE_NODE	NPN_DRIVE_BUS	correct
D7 (1N4148W)	PNP_PREDRIVE	PNP_DRIVE_BUS	PNP_PREDRIVE	correct

Table 5: Reverse V_{EBO} protection diodes. All five orientations verified correct against the netlist.

An earlier analysis, based on reading diode triangle glyphs in low-resolution KiCad footprint screenshots, concluded that D5 and D6 were reversed and blamed them for the bring-up fault. That conclusion was wrong and cost several hours. Physical cathode-band inspection contradicted it and the netlist settled it. The methodology consequence is recorded in Section 15.

4 Netlist Design Audit

A systematic review of `BJT_PUSH_PULL.net` produced eighteen findings. Severity is graded: **critical** means it prevents correct operation or destroys hardware; **high** means it degrades performance or reliability materially; **medium** means it should be corrected on the next revision; **note** means informational.

Four of these findings were subsequently confirmed on the bench as the causes of the bring-up failure (findings 3, 4, 5 and 6). The remainder are latent and define the next-revision requirements in Section 14.

4.1 Critical — feedback taken at the amplifier output, not at the load

U2.2 (–IN), U2.6 (OUT) and Q13.1 (B) are the same net, `/HV_DRIVE.NODE`. The output stage is entirely outside the amplifier’s feedback loop.

$$V_{\text{COIL_DRIVE}} \approx V_{\text{COMMAND}} - V_{BE}(\text{Q13}) - V_{BE}(\text{Q}_{\text{out}}) \approx V_{\text{COMMAND}} - 1.2 \text{ V} \quad (3)$$

Effects: an uncorrected DC offset of two base-emitter drops, approximately 1.2 V, which into 0.656Ω is roughly 1.8 A of error at zero command; crossover distortion uncorrected; V_{BE} thermal drift uncorrected, so the output wanders as the heatsink warms; output impedance not reduced by loop gain.

This was originally logged as the dominant defect with a recommended fix of moving the –IN tap to `COIL_DRIVE`. **That recommendation is withdrawn.** The current loop is closed externally by the SIM960 reading the DP50IP, so the offset, the crossover distortion and the thermal drift are all inside the external loop and are absorbed by the PID. The amplifier is intentionally a voltage buffer. The offset appears directly only in SIM960 manual mode, where a 0 V command produces approximately –1.2 V at the output, and that is what was seen throughout open-loop debugging.

The residual cost of the choice is real but bounded: the external loop must supply all the correction, so its gain determines the output impedance and the crossover linearity, and the amplifier contributes no local error reduction.

4.2 Critical — DP50IP transducer will be destroyed if the rails are raised

U1.10 is named `+15V` in the schematic symbol but is wired to `/+30V_0P`; U1.12 (`–15V`) is wired to `/–30V_0P`. Those nets are shared with the OPA455 supply pins.

The DP50IP absolute supply range is $\pm 14.25 \text{ V}$ to $\pm 15.75 \text{ V}$. The OPA455 is rated to $\pm 75 \text{ V}$, so the design clearly intends these rails to be raised above 15 V at some point. The moment they exceed approximately $\pm 15.75 \text{ V}$, a transducer costing several hundred dollars is destroyed. The only reason this has not happened is that bring-up has been conducted entirely at $\pm 15 \text{ V}$.

Fix: split the rails. Give U1 its own regulated $\pm 15 \text{ V}$ and run the amplifier section on whatever the required output swing demands. The nets must never be shared.

4.3 Critical — NT1 net tie has no footprint, so SGND and PGND are not bonded

The schematic does contain the intended single-point ground bridge, with NT1.1 on `/SGND` and NT1.2 on `/PGND`. The netlist shows:

```
(comp (ref "NT1") (value "NetTie_2") ... (field (name "Footprint") )    <- EMPTY
```

No footprint means nothing was placed on the PCB, which means the two grounds were never connected in copper. This is exactly the 200 k Ω measured between SGND and PGND during debugging, and it is why the amplifier section had no defined reference relative to the power section.

Fix: assign a `NetTie_2` footprint, or a zero-ohm link. Until respun, keep the external single-point bridge, at one point only.

4.4 High — E/D pin left floating

The netlist contains `unconnected-(U2-E_D-Pad8)`. TI's datasheet section 7.3.4 states that a floating E/D pin self-enables, but only via an internal 1 μ A source holding the pin approximately 2 V above E/D Com, against a disable threshold only 0.8 V away. TI explicitly warns that this arrangement is fragile.

It did not survive. This became bring-up fault 1, Section 7.

Permanent fix: TI recommends an external current source from V+ sufficient to hold the enable level above the shutdown threshold, plus a 30 pF capacitor from E/D to a low-impedance node for noise immunity. A stiff resistive divider also works and is what was fitted.

4.5 High — C22 hangs directly on the amplifier output with no isolation

C22.2 is on `/HV_DRIVE_NODE`, which is U2.6 (OUT). C22 is 1 μ F. The OPA455 capacitive load drive rating is 200 pF. This is a factor of 5000 over specification, and because R26 turned out to be Q12's base stopper rather than an output isolation resistor, there is nothing in series at all.

Effects: phase-margin collapse and probable high-frequency oscillation. This was masked until the E/D fault was cleared, because a disabled output stage cannot oscillate.

Fix on respin: a 10 Ω to 47 Ω series resistor between OUT and C22, or relocate C22 so it does not load the amplifier output directly.

A handheld multimeter in AC mode cannot detect this. The oscillation frequency is in the hundreds of kilohertz to megahertz range, well past the AC bandwidth of most handheld meters. An oscilloscope on `HV_DRIVE_NODE` is required.

4.6 High — V_{BE} multiplier divider current is too low relative to Q12 base current

Divider current is approximately

$$I_{div} \approx \frac{V_{spread} - V_{BE}}{R_2} \approx \frac{1.75 \text{ V}}{10 \text{ k}\Omega} \approx 0.18 \text{ mA} \quad (4)$$

Q12's collector current is set by R24: approximately 13 mA at ± 15 V, 27 mA at ± 30 V. With BD139 h_{FE} in the range 100 to 250, the base current is 0.05 mA to 0.28 mA.

The standard design rule is that divider current should be at least ten times the base current. Here the ratio is below two, and at ± 30 V it can invert entirely.

The immediate effect is that the bias spread becomes strongly dependent on Q12's h_{FE} , which varies between units and with temperature, so the trim drifts and cannot be set reproducibly. Simulation across the vendor model's h_{FE} range gave quiescent currents from 16 mA to 1.15 A, a factor of 70.

The deeper effect, which only emerged during the fix, is that the spread is not simply $M \cdot V_{BE}$ but

$$V_{spread} = M \cdot V_{BE}(Q12) + R_2 I_b(Q12) \quad (5)$$

and with $R_2 = 10 \text{ k}\Omega$ the second term alone contributes 0.5 V to 0.8 V. That term is what put a floor under the achievable quiescent current and made RV2 useless. See Section 7.2.

Fix: scale the divider down by approximately ten, for example $R_2 = 1\text{ k}\Omega$, $R_{23} = 220\text{ }\Omega$, $RV_2 = 500\text{ }\Omega$, giving approximately 1.8 mA of divider current. Keeping the same ratios preserves the trim range.

The trim *range* itself is adequate; an earlier suggestion to change R23 was wrong and is withdrawn:

$$M = 1 + \frac{R_2}{R_{23} + RV_2} = 1 + \frac{10\text{ k}\Omega}{2.2\text{ k}\Omega + 0\text{ k}\Omega \text{ to } 5\text{ k}\Omega} = 2.39 \dots 5.55 \quad (6)$$

$$V_{\text{spread}} = M \cdot V_{BE}(\text{Q12}) \approx 1.55\text{ V} \dots 3.6\text{ V} \quad (7)$$

against a target of approximately 2.4 V to 2.6 V for four junctions, which lands at $RV_2 \approx 1.1\text{ k}\Omega$ to $1.5\text{ k}\Omega$, about 25 % of travel.

4.7 Note — Q12 thermal coupling (retracted finding)

An earlier audit entry claimed Q12 was not thermally coupled to the output devices. The PCB file shows that Q12 is mounted in the heatsink row between Q10 and Q5 ($y \approx 90\text{ mm}$, the same row as the output devices), and the BOM marks its TO-126 footprint do-not-populate, meaning it is hand-mounted on the heatsink. Thermal tracking is designed in. The claim is retracted.

The real issue in this area is electrical rather than thermal. Q12's collector, emitter and base-divider connections run approximately 95 mm, 90 mm and 140 mm of 0.2 mm trace back to the drivers and divider, which is a large inductive loop in precisely the circuit that oscillates. On the next revision, keep Q12 on the heatsink but route the multiplier with short paired traces, or place it near the drivers with a separate thermal link.

4.8 High — COMMAND has no DC return

/COMMAND consists of J1.1, C21.1 (100 pF to SGND) and U2.3 (+IN). There is no resistor to ground. If J1 is ever unplugged, +IN floats on 30 pA of bias current and the output can wander anywhere, which with a ten-device output bank on $\pm 15\text{ V}$ rails and no current limiting is a hardware-destruction scenario.

This is required even though the command is set externally. The DC return does not define the signal; the SIM960 does that whenever it is connected. The resistor defines the node when the source is *not* connected: during power-up before the PID has settled, if a BNC is pulled, or if a cable fails. It costs nothing and its only effect during normal operation is to load the SIM960 output with 10 k Ω to 100 k Ω , which is negligible against the SIM960's output drive.

Fix: 10 k Ω to 100 k Ω from COMMAND to SGND.

4.9 High — Status Flag unconnected

The netlist contains `unconnected-(U2-Status_Flag-N-Pad5)`. Both overtemperature and overcurrent produce a high-impedance output on the OPA455, which is the same electrical signature as E/D shutdown. That ambiguity is exactly what slowed this debug: there was no way to distinguish "disabled" from "protected" without measuring the E/D pin directly.

Fix: 10 k Ω pull-up to 5 V referenced to E/D Com, plus a test point.

4.10 Medium — D1 and D2 are generic and under-specified

The value field is simply "D" with the description "DIODE GEN PURP 600V 15A T0220-2 (e.g., MUR1560G)". No part number is committed.

Two things need verifying: the repetitive surge rating against the 44 mJ per-shutdown flyback energy, and the recovery speed. D1 and D2 are also the only components defining COIL_DRIVE's potential when the output is unloaded.

4.11 Medium — no Zobel network or output snubber

Class-AB stages driving inductive loads commonly require a series R-C network at the output for high-frequency stability. /COIL_DRIVE contains only the ten ballast resistors, four diodes and connector J8. Nothing else.

Fix: add a Zobel network, for example $10\ \Omega$ in series with 100 nF to PGND, at COIL_DRIVE.

4.12 Medium — no DC bleed on COIL_DRIVE

With no coil connected, COIL_DRIVE floats. A high-value bleed, 10 k Ω to PGND, would define it for safe unloaded bring-up.

4.13 Medium — no output-stage current limiting

The output devices are protected only by the external supply's current limit. For a 10 A design driving ten paralleled power transistors, a single fault destroys the entire bank. Consider VI-limiter transistors across the drivers.

This is the single most consequential difference between this board and any monolithic alternative: the OPA549 has current limit, thermal shutdown and a status flag built in.

4.14 Medium — burden resistor at the transducer's maximum allowed value

R25 = 100 Ω . The DP50IP specification for measuring resistance R_M is 0 Ω to 100 Ω . Operating exactly at the maximum leaves no margin for tolerance or temperature.

At the wired 1:250 ratio, 10 A gives 40 mA of secondary current and 4.0 V across R25; 12.5 A gives 5.0 V.

4.15 Note — NPN devices use the PNP part's footprint library

MJL21194 (NPN) parts are assigned the footprint MJL21193G:T0545P2030X530X2900-3. Both are TO-264 and both are B-C-E, so this is very likely harmless, but it is a library cross-reference and deserves a one-time pin-map verification.

4.16 Note — RV2 is a 25-turn trimmer

RV2 is a Bourns 3296W. "Fully clockwise" requires twenty-five revolutions. The 3 k Ω maximum measured on a 5 k Ω part during bring-up suggests the end of travel was never actually reached. Verify the true endpoints with an ohmmeter rather than by feel.

4.17 Note — power dissipation

- 0.22 Ω ballasts at 2 A each (10 A divided by five) is 0.88 W each, in a TO-220-2 package, requiring heatsinking.
- R24 (1 k Ω axial) at ± 30 V is 0.76 W in an approximately 1 W part. Marginal; specify 2 W.
- 2.2 Ω base resistors (2512) at approximately 3.5 mW: fine.

- R1 ($100\,\Omega$) at approximately 14 mW: fine.

4.18 Critical — R24 is a passive pull-down, not a current source, and starves the PNP half

The lower half's entire drive comes from R24 ($1\,\text{k}\Omega$) to the negative rail, so its available sink current collapses as the output swings negative. Simulation shows the predrive spread falling from 2.2 V to 0.9 V with Q12 saturating, limiting negative full scale to roughly $-8\,\text{A}$ to $-9.5\,\text{A}$.

Lowering R24 to approximately $667\,\Omega$ restores $\pm 10.5\,\text{A}$, but every milliampere through R24 is supplied by the amplifier via Q12, and at $470\,\Omega$ the OPA455 reaches its 45 mA output limit. The proper fix is to replace R24 with an active current source.

This is a hard ceiling on the present board's negative full scale and has not been tested on the bench, because characterisation to date has been conducted at currents well below 1 A.

4.19 What the audit found correct

- All five protection diodes correctly oriented (Section 3.4).
- All ten base-resistor and ballast-resistor pairings correct and consistent.
- Five NPN and five PNP, balanced, with collectors correctly on their respective rails.
- Decoupling symmetric and generous: two $1000\,\mu\text{F}$ plus five $100\,\text{nF}$ per main rail.
- OPA455 PowerPAD correctly tied to $V-$.
- RV2 wiper tied to one end, the standard fail-safe against wiper wear.
- DP50IP wired for four turns, 1:250, 12.5 A nominal, appropriate for the 10 A target.

5 Bring-Up Debugging

The board did not work on first power-up. This section records every measurement and every physical modification in the order they were made, including the dead ends, because several of the dead ends are instructive and one of them produced a retracted conclusion that cost significant time.

5.1 Bench setup

Item	Configuration
Supplies	Four Delta ES 015-10, floating outputs, as two series pairs: main ± 15 V (PGND) and amplifier/sensor ± 15 V (SGND). All rails at ± 15 V; the design nets are named ± 30 V.
Ground	SGND to PGND bridged by one wire at the supply, single point. SIM960 BNC shells earth SGND.
Command	SIM960 in manual mode, 0.000 V, open loop.
Load	Ten 1 k Ω resistors in parallel (100 Ω) across J8–J9. No coil.
Instruments	Handheld multimeter; Rigol DS1104Z with $\times 10$ probe, single ground reference to PGND.

Table 6: Bench configuration for all bring-up debugging.

5.2 Phase A — board 1, initial symptom

#	Test or modification	Result	Conclusion
A1	All six supply leads from one bipolar supply, no load, no signal	Approximately 5 A; removing any single lead drops it to normal	Originally blamed on a ground loop. Actual cause was the disabled amplifier (B6). Moved to separate supplies.
A2	Separate supplies	Negative rail at limit (10 A) regardless of RV2 or command; positive rail alone draws nothing	Fault is on the PNP side
A3	Unpowered diode and resistance checks: PNP_PREDRIVE to PNP_DRIVE_BUS, PNP_PREDRIVE to -30 V, -30 V to PNP_DRIVE_BUS	0.5 V / 0.4 V / 0.25 V, same both directions	Resistor-network paths, not junction faults
A4	Q12 diode test	No shorts	Q12 healthy
A5	Output devices C-E, D1 and D2 positions, in circuit	Approximately 0.06 V both directions	Parallel paths; in-circuit measurement is ambiguous here
A6	Modification: D2 lifted and tested isolated	Open one way, 0.3 V the other	D2 healthy
A7	Output device pinout against board nets	Pad 1 to 2.2 Ω to drive bus (B), pad 2 to rail (C), pad 3 to 0.22 Ω to COIL_DRIVE (E)	Matches schematic

Table 7: Phase A: board 1 initial investigation.

5.3 Phase B — board 2 and the amplifier fault

A second board was assembled independently, with all components verified before fitting. It reproduced the symptom identically, which established that the cause was design-level rather than

component damage. That single observation is what redirected the investigation away from component testing.

#	Test or modification	Result	Conclusion
B1	Same power-up as board 1	Identical symptom	Design-level cause
B2	RV2 resistance	Pins 2–3 shorted; pin 1–3 sweeps $0.6\ \Omega$ to $3000\ \Omega$. 25-turn, clockwise is maximum resistance	Pins 2–3 is an intentional wiper-to-end fail-safe tie; pot healthy
B3	Diode cathode bands inspected	All five correct	Diodes correct; later confirmed by netlist
B4	Powered, 0 V command	NPN to PNP predrive $0.67\ \text{V}$; $R_2 = 10\ \text{k}\Omega$; Q12 $V_{CE}\ 0.67\ \text{V}$, $V_{BE}\ 0.12\ \text{V}$, $V_{BC}\ -0.56\ \text{V}$; R24 drop $0.4\ \text{V}$; R23 drop $0.12\ \text{V}$	Both predrive nodes collapsed toward the negative rail; Q12 off and innocent
B5	Rails reduced to $\pm 15\ \text{V}$, amplifier pins measured	+IN $0\ \text{V}$, –IN = OUT = $-15\ \text{V}$	Amplifier output railed negative
B6	E/D (pin 8) relative to E/D Com, signal rails only	$0.06\ \text{V}$ against a disable window of $0\ \text{V}$ to $0.35\ \text{V}$	Output stage disabled
B7	Command +IN = $4\ \text{V}$	–IN = OUT = $2.5\ \text{V}$	Textbook disabled signature: $160\ \text{k}\Omega$ output impedance dividing against the external network
B8	Modification 1: $10\ \text{k}\Omega$ from +15V_OP to pin 8, $4.7\ \text{k}\Omega$ pin 8 to SGND, giving approximately $4.8\ \text{V}$ on E/D	+IN $4\ \text{V}$ gives –IN $4\ \text{V}$ gives OUT $4\ \text{V}$	Fault 1 fixed

Table 8: Phase B: board 2, isolating the operational amplifier fault.

The B7 result deserves emphasis because it is the diagnostic that identified the fault unambiguously. A dead amplifier gives a rail. An enabled unity follower gives a clean $4\ \text{V}$. An amplifier whose *output stage* is disabled while its inputs remain active gives an attenuated but tracking voltage, because the output impedance rises to approximately $160\ \text{k}\Omega$ and the signal divides against the external network ($R_2 + R_{23} + RV_2 + R_{24} \approx 13.5\ \text{k}\Omega$). The measured $2.5\ \text{V}$ from a $4\ \text{V}$ input is that divider.

5.4 Phase C — main rails on after the E/D fix

Measurement note. The ballast readings over-counted: $13\ \text{mV} \times 5/0.22\ \Omega = 295\ \text{mA}$ against $170\ \text{mA}$ actually delivered by the supply. Approximately $10\ \text{mV}$ across $0.22\ \Omega$ is at the handheld multimeter’s noise floor. Use the supply current meters to set bias, not ballast millivolts. Furthermore, all DC readings in this phase were taken while the stage was oscillating (Phase E), so every one of them is the average of a switching waveform rather than a static operating point.

The C18 observation. Equal positive and negative supply currents with only $0.1\ \text{A}$ flowing and a $100\ \Omega$ load cannot be load current: a load current would appear on one rail and return through PGND. Equal currents on both rails mean current flowing from the positive rail through the output stage to the negative rail, that is, shoot-through. Combined with C4 (no shoot-through through R1) this located the conduction in the output banks themselves, which is what oscillation produces.

#	Test or modification	Result	Conclusion
C1	Main rails on, 0.2 A limit	Negative rail still at limit; PNP_DRIVE_BUS minus HV fixed at 0.54 V regardless of RV2; HV tracks command	Amplifier now works; fault is downstream
C2	−15V rail under limit	−1.2 V	Supply in constant-current mode
C3	Unpowered resistance: PNP_DRIVE_BUS to −15V; COIL_DRIVE to +15V; COIL_DRIVE to −15V	0.5 M Ω ; 800 k Ω ; 80 k Ω	No static shorts
C4	Voltage across R1 (100 Ω)	0.389 V = 3.9 mA	No driver shoot-through through R1
C5	Ballast resistors (0.22 Ω)	Three of five PNP at 24 mV; NPN zero	PNP bank is conducting
C6	Limit raised to 4 A for 5 s	Rail stays at approximately −1.2 V	Voltage constant across a 20-fold current change implies junction clamping, not a resistive short
C7	Supply terminal against board pad	No drop	Not a cable or connector issue
C8	SGND to PGND resistance	200 kΩ	Grounds never bonded; NT1 net tie has no footprint
C9	Modification 2: SGND bridged to PGND at the supply	All five PNP ballasts 13 mV; rail −1.54 V at 0.17 A; HV to PNP_PREDRIVE 1.7 V	Partial improvement
C10	Limit 2 A for 5 s	Rail does not rise	Still clamped

Table 9: Phase C, part 1: main rails energised after the E/D fix.

5.5 Phase D — netlist audit

At this point in-circuit probing had been exhausted and the netlist was obtained. One parse answered questions that days of probing could not, and produced the eighteen findings in Section 4, including three latent bugs unrelated to the fault being chased. The relevant findings for the immediate fault were: no separate NPN_PREDRIVE net; a 1.2 V output offset by design; NT1 unplaced; E/D and Status Flag unconnected; C22 directly on the amplifier output; Q13 and Q14 without base stoppers; and multiplier divider current below Q12’s base current.

5.6 Phase E — oscillation

Mechanism. An emitter follower driving a capacitive load presents a negative real part at its input over a band of frequencies. The driver transistor’s base sees the output bank’s reflected input capacitance, C_{load}/β plus the bank’s own C_{be} , through the base circuit’s inductance. That inductance, that capacitance and the negative input resistance form a Colpitts oscillator. Five paralleled TO-264 devices present a large C_{be} , and the PCB traces to the drivers supply the inductance. With nothing in the base circuit to dissipate energy, the loop oscillates.

The two observed modes had distinct causes. The 1.67 MHz mode closed through C22, which is why lifting C22 removed it. The 3.09 MHz mode is the driver-stage Colpitts loop proper and required base stoppers.

Why the oscillation explains the primary symptom. A stage oscillating at several megahertz with volts of amplitude on both drive buses drives both output banks into conduction on every cycle. Current therefore flows from the positive rail through the NPN bank, into the PNP bank and to the

#	Test or modification	Result	Conclusion
C11	Unpowered COIL_DRIVE to PGND; COIL_DRIVE to -30V	Ramps then open; $0.7\text{ V} / 0.4\text{ V}$	Bulk-capacitor charging and parallel paths; benign
C12	$2\ \Omega$ load at 0.17 A limit	No change	Test ineffective at that current
C13	Modification 3: D2 lead cut	Unchanged (-1.73 V at 0.17 A); COIL_DRIVE exactly at PGND	D2 eliminated as a cause
C14	COIL_DRIVE to PGND resistance, unpowered	$460\text{ k}\Omega$	No board short. The test load J8 to J9 to DP50IP to PGND was effectively a short circuit
C15	Modification 4: load changed to ten $1\text{ k}\Omega$ in parallel ($100\ \Omega$)	—	Realistic, survivable test load
C16	0 V command, 0.1 A limit	Negative rail -2.77 V ; PNP ballasts $11/8/6\text{ mV}$ (two off); NPN approximately zero (one at 0.032 mV)	PNP-heavy conduction
C17	Node voltages relative to COIL_DRIVE	HV $+0.960\text{ V}$; PNP_PREDRIVE -0.875 V ; NPN_DRIVE_BUS $+0.60\text{ V}$; PNP_DRIVE_BUS -0.58 V ; HV to PNP_PREDRIVE 2.0 V	Both banks biased near or at turn-on
C18	Supply meters	$+15\text{ V } 0.1\text{ A}$ (CV), negative $2.27\text{ V } 0.1\text{ A}$ (CC)	Equal currents imply rail-to-rail current, not load current
C19	RV2 full sweep, watching PNP_DRIVE_BUS	-0.58 V to -0.59 V	Approximately 1.5-fold current change; V_{BE} is logarithmic. RV2 works; wrong observable

Table 10: Phase C, part 2: continued.

negative rail, bypassing the load. That is why both supplies sat at their current limits with a 0 V command and no load current, and why raising the limit simply raised the current. It also explains why RV2 appeared to have no effect: the bias trim sets the quiescent point of a stage that was not in a quiescent state.

#	Test or modification	Result	Conclusion
E1	HV_DRIVE_NODE to PGND, DC, 0 V command	0.1 V, wandering up to approximately 1 V	Oscillation suspected
E2	Oscilloscope, first capture	36 ns, 1.32 V peak-to-peak "square"	Undersampled; discarded
E3	Oscilloscope, proper capture, all nodes against PGND	All nodes at 600 ns period (1.67 MHz) with an approximately 270 ns ring (3.7 MHz); fast-ramp, sharp-fall, decaying-ring shape. NPN_DRIVE_BUS 3.18 V peak-to-peak, symmetric about zero; PNP_DRIVE_BUS 2.36 V peak-to-peak, 0 V to -2.36 V; COIL_DRIVE 2.52 V peak-to-peak, about two-thirds below zero with a crossover double-peak; HV 2.70 V peak-to-peak, -0.6 V to +2.1 V	The energy source is downstream of the amplifier: driver emitter swing exceeds amplifier output swing
E4	Modification 5: C22 lifted at one end	1.67 MHz mode gone. All nodes at 324 ns (3.09 MHz). HV approximately 1.53 V peak-to-peak; NPN_DRIVE_BUS 1.28 V peak-to-peak. Supplies 50 mA each, at limit	C22 closed the 1.67 MHz loop. A second mode remains
E5	Limits raised to matched 0.5 A	Still at limit; +15V holds, negative rail -3 V	Current fills any limit, confirming oscillation-driven hard conduction

Table 11: Phase E: identification of the oscillation.

6 SPICE Simulation Programme

Bench probing had identified an oscillation but could not determine the value of base stopper required, whether C22 had to stay lifted, or whether the bias trim could reach a useful operating point. Those three questions were answered in simulation and then confirmed on the bench. This section records the methodology, the results, and a direct comparison of prediction against outcome.

6.1 Methodology

The circuit was built directly from the netlist connectivity rather than redrawn, so the simulated topology is the fabricated topology. Simulation used ngspice.

Devices

Two model sets were used. The first was hand-parameterised from datasheet values, with $T_F = 1/(2\pi f_T)$ and junction capacitances from the datasheet curves: BD139 $f_T \approx 150$ MHz, MJE1503x ≈ 30 MHz, MJL2119x ≈ 4 MHz. The second used onsemi vendor .LIB models for all five transistor types (BD139, MJE15032, MJE15033, MJL21194, MJL21193). Results from the vendor set supersede the hand-parameterised set wherever they differ.

Operational amplifier

Two representations. A macromodel with $A_0 = 135$ dB, gain-bandwidth 6.5 MHz, a second pole near 20 MHz, open-loop output impedance $90\ \Omega$ and a tanh rail clamp; and the TI OPA455 PSpice macromodel (SBOMB81B). The macromodel converges far more readily and was used for parameter sweeps; the TI model was used for validation.

Parasitics

Trace inductances were derived from the .kicad_pcb file rather than guessed. Amplifier to Q13 approximately 40 nH; Q13 to Q12 approximately 95 nH; Q12 emitter to PNP_PREDRIVE approximately 90 nH; divider approximately 140 nH; base and emitter leads of each output

device 10 nH. Supply wiring 0.5 μ H damped by 10 Ω ; 1000 μ F bulk capacitors modelled with 30 m Ω ESR and 20 nH ESL; 100 nF decoupling with 2 nH. A `pscale` factor allowed all parasitics to be scaled together to test robustness.

Loads

The 100 Ω resistor bank (with 300 nH of splay inductance), the real coil (0.886 mH, 0.656 Ω), a 5 m Ω short, and a 10 μ H stand-in for the coil. The stand-in presents the same effective open circuit at megahertz frequencies but settles in approximately 75 μ s instead of milliseconds, which makes stability sweeps tractable. It was validated against the real coil in every case where both converged.

Excitation

Stability was probed by injecting a current pulse at the output node rather than stepping the command, because a command step excites the amplifier’s own dynamics and confounds the result. Edges were softened to 200 ns (from an initial 5 ns) to avoid numerical artefacts being mistaken for physical ringing.

6.1.1 Convergence

ngspice failed to find an operating point in the as-built configuration, which is expected: a class-AB stage with ten paralleled devices, an oscillating loop and no local feedback has no well-conditioned DC solution. Three measures made the simulations run.

1. `.nodeset` seeding, with node voltages taken from the operating point of the macromodel version of the same circuit. This is the single most effective measure.
2. Softened excitation edges, as above.
3. The 10 μ H load stand-in, which removes the millisecond time constant from the transient.

Adding `gminsteps=100 srcsteps=100` *broke* cases that had previously converged and was reverted. This is recorded because it is a natural thing to try. Solver options were otherwise `method=gear maxord=2 reltol=1e-3 abstol=1e-10 vntol=1e-6 trtol=7 rshunt=1e10`.

6.2 Phase F — estimated models

The first simulation pass reproduced four qualitative bench observations without being tuned to do so: the as-built oscillation, its persistence after C22 was lifted, the -1.2 V output offset, and the negative rail carrying more current than the positive.

Configuration	Result
As-built, or C22 lifted, no stoppers	Oscillates
C22 lifted plus 22 Ω to 33 Ω stoppers	Oscillates
C22 lifted plus 47 Ω	Stable, poor margin
C22 lifted plus 100 Ω	Stable, good margin
No stoppers, real coil	Oscillates

Table 12: Phase F stopper sweep with hand-parameterised models. The 100 Ω result held at bias currents from 16 mA to 1.15 A, with parasitics tripled, into the real coil, and into a short.

The last row is important on its own: the coil does not stabilise the stage. At megahertz frequencies an 0.886 mH inductor is an open circuit, so the oscillation loop does not see the load at all. Any hope that connecting the real coil would damp the problem was therefore misplaced.

Phase F also gave the drive cost of the stoppers (approximately 3%, so 10 A requires about 8.9 V of command) and a crossover gain dip ranging from 5% at 300 mA per bank to 90% at 2 mA per bank.

6.3 Phase G — vendor models, validation and the revised recommendation

Phase G repeated the work with onsemi vendor models and the TI OPA455 macromodel. It changed the recommendation.

6.3.1 Validation against the bench

Case	Simulated	Bench
As-built, C22 fitted	3.73 MHz	3.7 MHz ring
C22 lifted, no stoppers	3.20 MHz, 1.58 V pk-pk	3.09 MHz, approximately 1.53 V pk-pk

Table 13: Simulation against measurement. Agreement is 4% in frequency and 3% in amplitude. No parameter was fitted to the bench data.

This is the result that justified trusting the model’s predictions. The frequency and amplitude of a parasitic oscillation depend on device capacitances, transit times and trace inductances in combination, so agreement at this level across two configurations indicates the parasitic extraction and the device models are both close to right.

The 1.67 MHz sawtooth envelope was never reproduced. The most likely explanation is an interaction with the supplies’ constant-current mode, which is not modelled: the supplies were in current limit throughout the bench observation, and a current-limited supply behaves as a nonlinear source with its own control loop. This remains the one unexplained bench observation from Phase E.

6.3.2 Stopper sweep across the full command range

Q13 / Q14	−6 V	−3 V	−1 V	0	+1.5 V	+3 V	+6 V	+10 V
100 Ω / 100 Ω	ok	osc	osc	ok	ok	ok	ok	ok
200 Ω / 200 Ω	ok	ok	ok	ok	ok	ok	ok	ok

Table 14: Stability against command voltage into the real coil, vendor models. The 100 Ω recommendation from Phase F is adequate only near zero current; with vendor models the PNP half oscillates at −1 V and −3 V commands.

The 200 Ω configuration also held with parasitics doubled, at both bias extremes, into a short and into the real coil. 220 Ω was verified separately because that was the value available, and it was stable at every command point.

6.3.3 Bias, deadband and the trimmer

Phase G also resolved why RV2 appeared to have no authority, which had been misdiagnosed twice. Quiescent current proved extremely sensitive to Q12’s h_{FE} : across the vendor model range it varied from 16 mA to 1.15 A per bank. This is audit finding 6 (Section 4.6) made visible.

More importantly, the simulation established that *no change at the trimmer can reach a deadband*. From Equation 5, the spread has two terms. Raising R23, or adding series resistance at RV2, shrinks $M \cdot V_{BE}$ but *increases* $R_2 I_b$, because less divider current makes Q12’s base current relatively more significant. Simulation showed that even 47 k Ω in series with RV2 bottoms out at 16 mA per bank

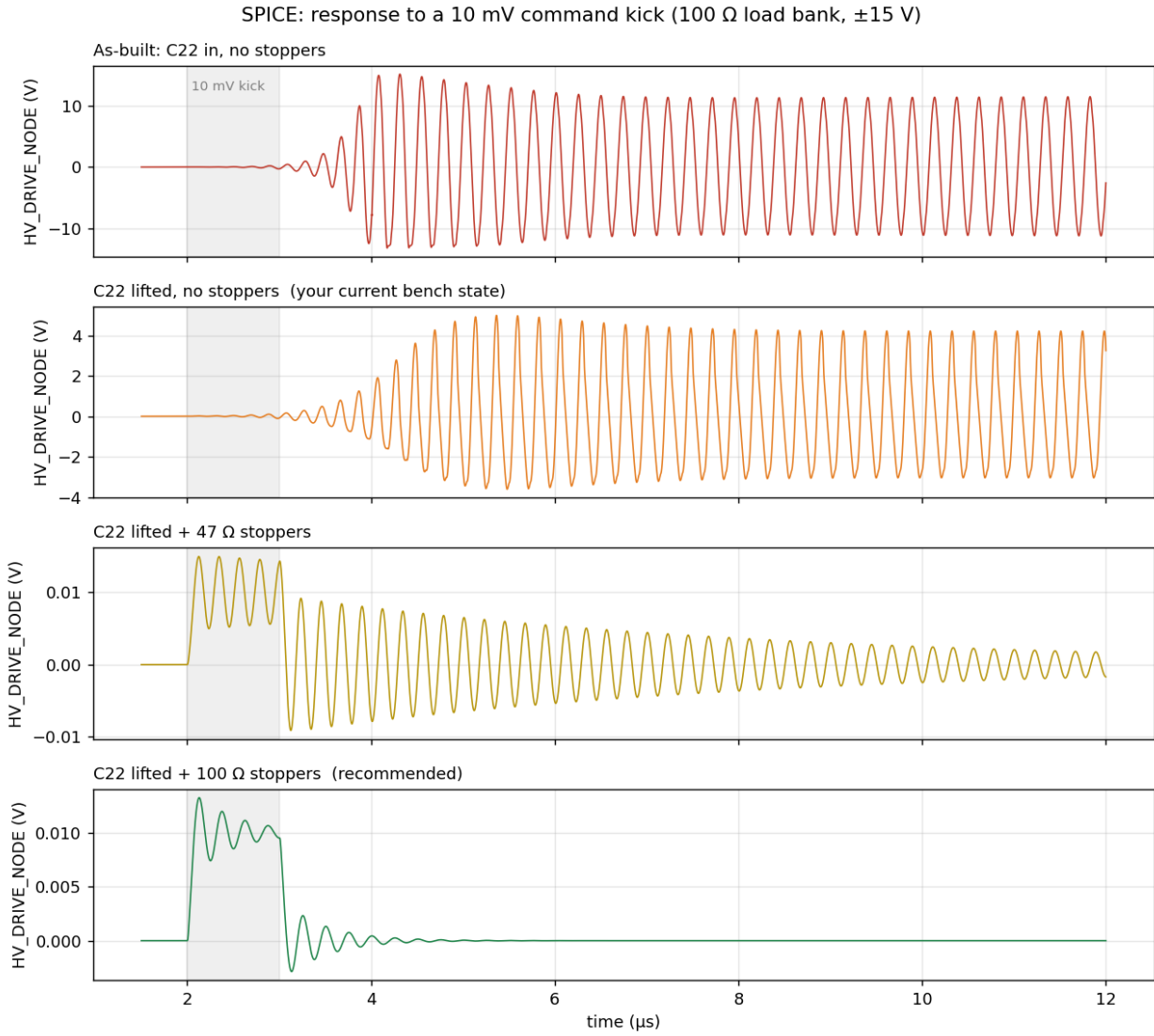


Figure 1: Simulated oscillation across stopper configurations. The as-built and no-stopper cases sustain oscillation indefinitely; increasing the stopper value converts the sustained oscillation into a damped ring-down.

with no deadband reachable. Only lowering R_2 shrinks both terms at once. A 10 k Ω resistor across R_2 was shown to lower the floor enough to reach an adjustable deadband of 200 mV to 315 mV of command.

This prediction mattered because the first physical attempt had been made at the trimmer, and it failed exactly as the simulation said it would.

6.3.4 Negative full-scale limit

R_{24} at 1 k Ω starves the PNP half beyond approximately -8 A: the predrive spread collapses from 2.2 V to 0.9 V with Q12 saturating. Approximately 667 Ω restores ± 10.5 A while keeping the OPA455 below its 45 mA output limit. This became audit finding 18 and remains untested on the bench.

6.3.5 Model caveat

Three of the five vendor models have $NF < 1$, which is physically impossible and is a curve-fitting artefact. Absolute quiescent-current and deadband numbers from those models are directional only.

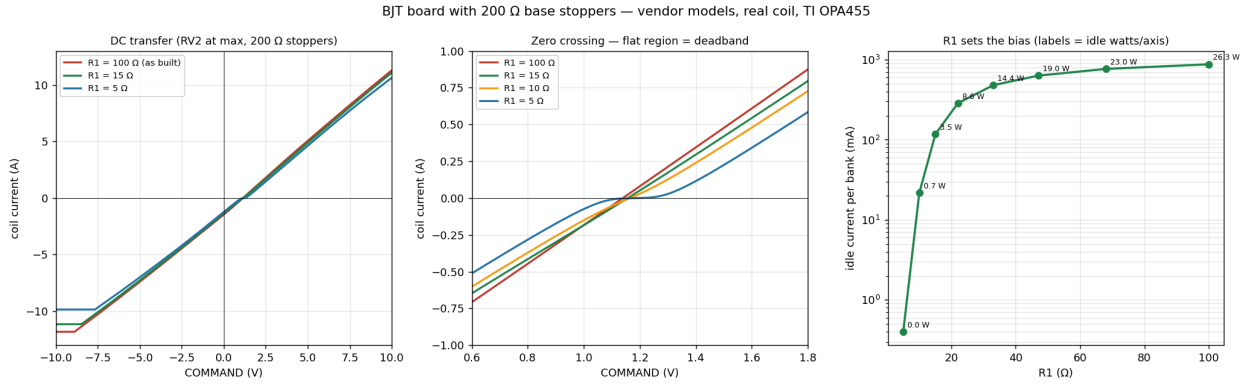


Figure 2: Simulated DC transfer characteristic (left), zero-crossing detail showing the crossover region (centre), and idle current against the value placed across R2 (right). The right-hand panel is the basis for the 15.6 k Ω value actually fitted.

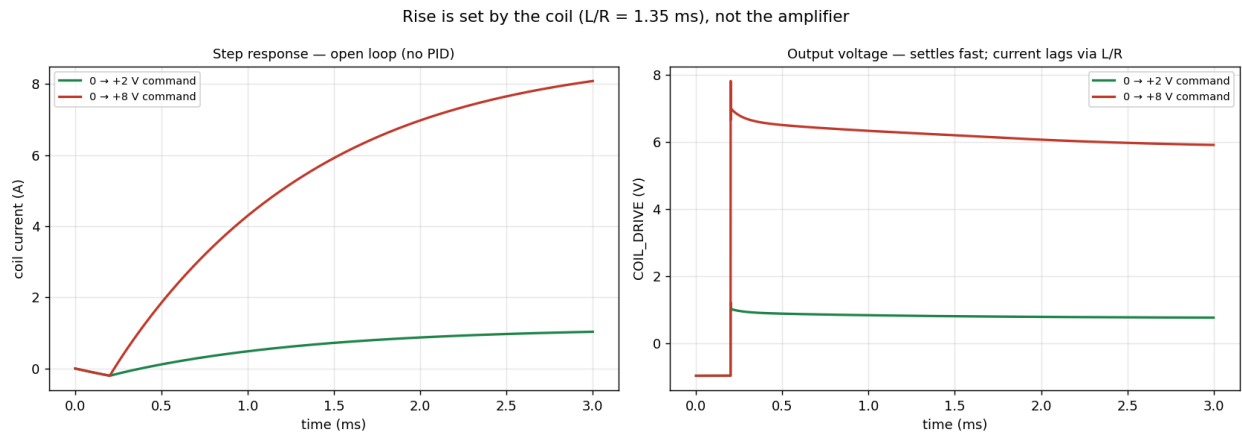


Figure 3: Simulated step response into the real coil with 200 Ω stoppers. The millisecond-scale settling is the L/R time constant, not an amplifier limitation.

The stability conclusions do not depend on NF and are unaffected.

6.4 Predictions against outcomes

The following table is the record of what the simulation programme predicted and what the bench subsequently showed. It is the basis for treating the model as usable for the next revision.

On the surviving ring. The 3.7 MHz ring observed in characterisation test 7 is the same loop that previously oscillated. Base stoppers do not remove the resonance; they add loss to it. The loop's negative resistance is now smaller in magnitude than the stopper's positive resistance, so the pole pair sits in the left half-plane and the response is a decaying ring rather than a sustained oscillation. Three cycles at 30 % overshoot corresponds to a Q of roughly 2.5, which is a normal margin for this kind of stage. If greater margin is wanted the options are a larger stopper (at the cost of drive headroom), a Zobel network at the output, or ferrite beads on the driver bases.

Prediction	Bench outcome	Verdict
3.20 MHz at 1.58 V pk-pk with C22 lifted and no stoppers	3.09 MHz at approximately 1.53 V pk-pk	Confirmed, within 4 % and 3 %
3.73 MHz as-built	3.7 MHz ring	Confirmed
Stoppers of at least $200\ \Omega$ stabilise the full command range	$220\ \Omega$ stable at every command point tested	Confirmed
C22 may remain fitted once stoppers are at least $200\ \Omega$	C22 left fitted, no oscillation	Confirmed
The coil will not stabilise the stage	Not tested directly; the resistor bank and the coil both oscillated before the fix	Consistent
RV2 cannot reach a deadband while $R_2 = 10\ \text{k}\Omega$	Trimmer had no useful authority; best reachable was 875 mA per bank	Confirmed
Adding series resistance at the trimmer cannot fix it	Attempted first on the bench; it failed	Confirmed
Lowering R_2 is the only route to a low quiescent point	$15.6\ \text{k}\Omega$ across R2 gave a trimmable 10 mA	Confirmed
Approximately 4 MHz damped ring-down survives with stoppers fitted	3.70 MHz, 30 % overshoot, three cycles, 1.1 μs settling	Confirmed
1.67 MHz sawtooth mode	Present on the bench, never reproduced in simulation	Not reproduced; attributed to supply constant-current interaction
R24 starves the PNP half beyond approximately $-8\ \text{A}$	Not yet tested; all characterisation below 1 A	Open
Quiescent current varies 70-fold with Q12 h_{FE}	Consistent with the difficulty of setting bias reproducibly	Consistent, not quantified

Table 15: Simulation predictions against bench outcomes. Nine confirmed, one not reproduced, one open, two consistent but not independently quantified.

7 The Fix

Four independent faults had to be cleared before the board worked. They are listed in the order they were found. Each was confirmed on the bench.

7.1 Base stopper installation

The stoppers had to be fitted to a fabricated board with no provision for them, so they were installed by cutting PCB traces. The important constraint is that *the stopper must carry only base current*. If it is placed so that other currents pass through it, it becomes a series element in the bias network and shifts the operating point.

For each driver this requires two cuts and one jumper. Q13's base node connects both to the amplifier output and to Q12's collector and R2. Q14's base node connects both to Q12's emitter and to R24 and RV2. In each case the stopper goes between the transistor's base pad and the rest of the node, with Q12, R2, R24 and RV2 on the far side of it.

7.2 Why the bias fix had to be at R2

The V_{BE} multiplier spread is given by Equation 5:

#	Fault	Root cause	Fix applied
1	Amplifier output disabled	OPA455 E/D pin left unconnected in the schematic; measured 0.06 V above E/D Com, inside the 0 V to 0.35 V disable window. Output impedance approximately 160 k Ω	10 k Ω from +15V_OP to pin 8, 4.7 k Ω from pin 8 to SGND, giving approximately 4.8 V on E/D
2	Amplifier had no reference relative to the power section	NT1 net tie had no footprint assigned, so SGND and PGND were never bonded in copper; measured 200 k Ω apart	Single external bridge wire at the supply
3	Parasitic oscillation near 3 MHz, PNP side dominant	Q13 and Q14 emitter followers driving the reflected input capacitance of the five-device output banks with no base stoppers. The amplifier also saw that capacitance with no isolation, via C22	220 Ω base stoppers at Q13 and Q14, fitted via PCB trace cuts
4	Rail-to-rail quiescent current; RV2 had no useful authority	V_{BE} -multiplier floor too high. With $R_2 = 10$ k Ω the $R_2 I_b$ term of Equation 5 alone held 0.5 V to 0.8 V. RV2 could reach no lower than 875 mA per bank, or 26 W of idle dissipation	10 k Ω plus 5.6 k Ω in series (15.6 k Ω) tacked across R2, giving $R_{2,\text{eff}} \approx 5.86$ k Ω

Table 16: The four faults and their fixes. All four were required; clearing any three left the board non-functional.

$$V_{\text{spread}} = \underbrace{\left(1 + \frac{R_2}{R_{23} + RV_2}\right)}_{\text{trimmable}} V_{BE}(\text{Q12}) + \underbrace{R_2 I_b(\text{Q12})}_{\text{floor}} \quad (5)$$

The first term is what the trimmer controls and what textbook treatments describe. The second term is normally negligible because the divider current is designed to swamp the base current. Here it is not: the divider runs at 0.18 mA against a base current of 0.05 mA to 0.28 mA (audit finding 6), so $R_2 I_b$ with $R_2 = 10$ k Ω contributes 0.5 V to 0.8 V regardless of the trimmer setting.

Raising R23, or adding series resistance at RV2, reduces the first term but *increases* the second, because reducing divider current makes the base current relatively larger. The two effects partly cancel, which is why the first physical attempt, made at the trimmer, produced no useful change. Lowering R_2 reduces both terms simultaneously: it reduces M directly and it reduces the $R_2 I_b$ product directly.

The value fitted was 15.6 k Ω across the existing 10 k Ω , giving $R_{2,\text{eff}} = 5.86$ k Ω .

7.3 Result

No oscillation at HV_DRIVE_NODE. Quiescent current trimmable with RV2, set to 10 mA cold and settling to 50 mA warm. This is just into class-AB, approximately 1.5 W of idle dissipation per axis against 26 W before.

Two changes that had been expected were *not* required. C22 stays fitted: with 220 Ω stoppers it is stable at every command point, and the earlier lift was necessary only at 100 Ω . R1 stays at 100 Ω . D2 was restored.

7.4 As-built configuration

An earlier revision of the debug log recorded a 200 Ω burden giving 0.8 V A⁻¹. That was incorrect; the fitted burden is 100 Ω and the scale factor is 0.4 V A⁻¹. All characterisation figures in Section 8 use 0.4 V A⁻¹.

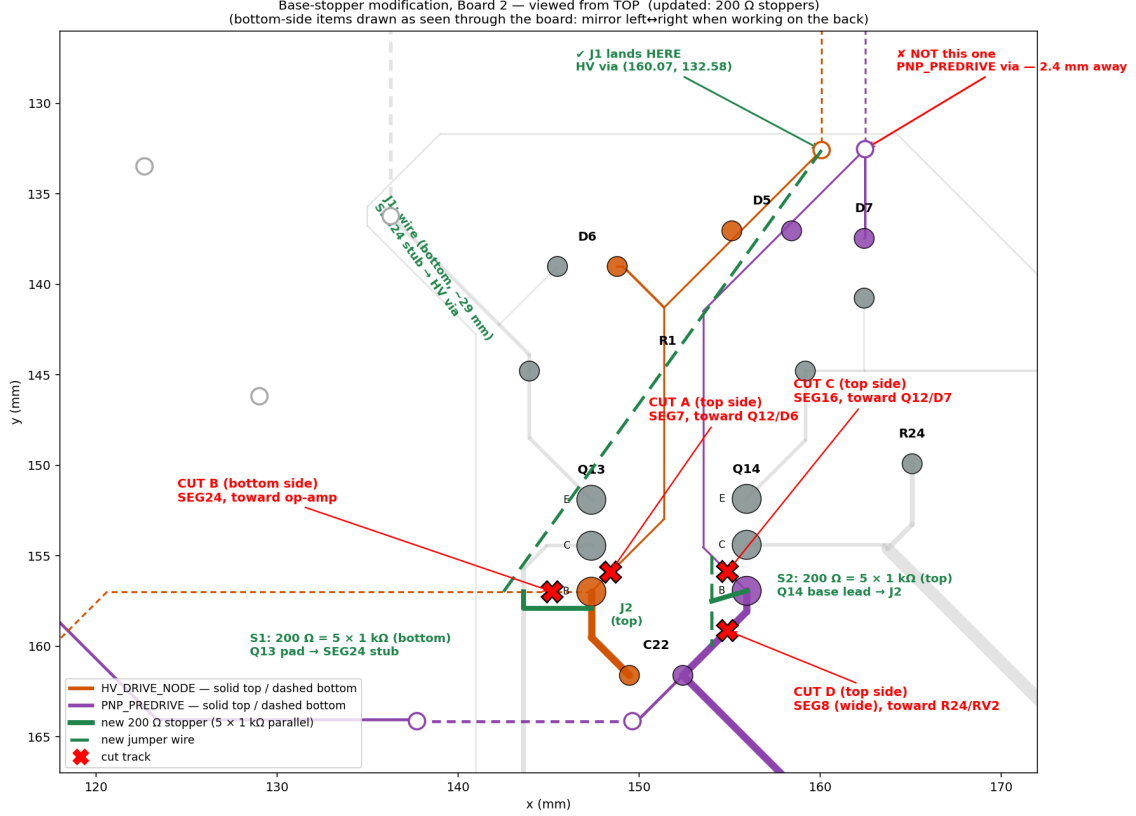


Figure 4: Trace-cut plan for the base stoppers. The drawing is labelled $200\,\Omega$; $220\,\Omega$ was fitted, being the value available, and was verified stable in simulation at every command point. Leads were kept under $5\,\text{mm}$, since lead inductance in series with the stopper partially defeats it.

8 Characterisation Results

Board 1V1, 25 September 2026. Load: $113.2\,\Omega$ non-inductive. Command from a function generator; monitor from the DP50IP BNC at $0.4\,\text{V A}^{-1}$. Supply limits matched. Oscilloscope Rigol DS1104Z, CH1 monitor (referred to below as the monitor channel), CH2 command.

The load was deliberately resistive rather than the real coil. A resistive load removes the L/R time constant and the coil's self-capacitance from the measurement, so the amplifier's own behaviour is visible. It also keeps the current low enough to be safe on an unprotected output stage.

8.1 Quiescent current against temperature

8.2 Transfer characteristic

8.2.1 Extracting the signal from the noise

Peak-to-peak readings on the monitor channel include the noise floor and are unusable for gain extraction. Variance is the reliable metric, because for a triangle wave

$$\sigma^2 = \frac{V_{pp}^2}{12} \quad (8)$$

This formula reproduces the command channel's $7.78\,\text{V}$ amplitude exactly, which validates the method. Applying it to the monitor channel and subtracting the noise variance gives approximately

Item	Value	Note
Q13, Q14 base stoppers	220 Ω each	one per driver, between pad and node, not one per cut
Across R2 (10 k Ω , 1206)	10 k Ω + 5.6 k Ω in series = 15.6 k Ω	$R_{2,\text{eff}} \approx 5.86 \text{ k}\Omega$
E/D divider	10 k Ω / 4.7 k Ω	approximately 4.8 V on pin 8
SGND to PGND	external bridge, single point	at the supply
C22	fitted	do not lift
R1	100 Ω	unchanged
D1, D2, D3	fitted	D2 was cut during debugging and has been restored; D3 also refitted
RV2	trimmed for 50 mA warm	re-check hot after any change
Rails	$\pm 15 \text{ V}$, matched current limits	no on-board current limiting exists
Current monitor	DP50IP, four turns (1:250), 100 Ω burden	0.4 V A $^{-1}$, equivalently 2.5 A V $^{-1}$

Table 17: Board 1V1 as-built configuration after the fix.

Time from cold	Both-supply current	Note
0 min	10 mA	trimmed here
15 min	50 mA	settled, stable

Table 18: Quiescent drift from cold. The fivefold rise is the expected V_{BE} tempco acting against an imperfectly tracking multiplier; it settles and does not run away, which is the criterion that matters. Trim hot.

29 mV to 30 mV pk-pk at 10 Hz and 100 Hz.

8.2.2 Gain

At 0.4 V A $^{-1}$, an ideal 70.7 mA pk-pk produces 28.3 mV pk-pk on the monitor. The extracted value is 29 mV to 30 mV. The incremental gain is therefore within approximately 2 % of the load-resistance prediction, in phase, and flat from 10 Hz to 1 kHz.

Correction to an earlier analysis. A first pass over this data concluded the gain was 6.5 times low. That conclusion used 0.4 A V $^{-1}$ where the transducer scale is 0.4 V A $^{-1}$, understating every current by a factor of $1/0.4^2 = 6.25$. With the correct scale the gain is as designed. The error is recorded because the wrong figure appeared plausible for several minutes and drove a search for a nonexistent fault.

8.2.3 Offset

A DC command of 10 V produced 0.07 A through the 113.2 Ω load, that is 7.92 V at COIL_DRIVE, implying an offset of approximately 2.1 V at unity slope. The nominal offset is two base-emitter drops, approximately 1.2 V. The additional 0.9 V is not explained by a single-point measurement and requires a two-point DC sweep to separate slope from offset properly. Candidate contributions are the base stopper drops, the finite bias spread with the multiplier trimmed near deadband, and the driver V_{BE} at load.

This offset is not an error in closed-loop operation: the external SIM960 integrator drives it to zero. It matters only in manual mode and for the available output swing, which it reduces by approximately 2 V on the positive side.

#	Command	Monitor measured	Current	Ideal
0	none	16.8 mV pk-pk noise, centre approximately -3 mV; counter reads 79.4 Hz	42 mA pk-pk noise, approximately -7.5 mA offset	0
1	10 Hz triangle, ± 1 V	17.2 mV pk-pk, essentially the noise floor	not resolvable	17.7 mA pk-pk
2	10 Hz triangle, ± 4 V	42.0 mV pk-pk including noise; AC-RMS 9.1 mV; variance $83.5 \mu\text{V}^2$	72 mA pk-pk	70.7 mA pk-pk
3	100 Hz triangle, ± 4 V	44.8 mV pk-pk; amplitude 25.7 mV; variance $89.3 \mu\text{V}^2$	72 mA pk-pk, no visible phase lag	70.7 mA pk-pk
4	1 kHz triangle, ± 4 V	46.4 mV pk-pk; amplitude 25.7 mV; variance $124 \mu\text{V}^2$	72 mA pk-pk; lag under $20 \mu\text{s}$, under 7°	70.7 mA pk-pk
5	10 Hz square, ± 0.5 V	step approximately 2 mV to 3 mV	approximately 6 mA	8.8 mA
6	10 Hz square, 0 V to 4 V	step 12.4 mV; damped ringing near 4 MHz, second burst approximately $3.5 \mu\text{s}$ later	31 mA	35.3 mA
7	10 Hz square, ± 2 V	step 12 mV to 14 mV; approximately 30 % overshoot; 3.70 MHz ring, 270 ns period; settles in approximately $1.1 \mu\text{s}$; three visible cycles	31 mA	35.3 mA

Table 19: Transfer measurements, 25 September 2026. Ideal means $V_{\text{cmd}}/113.2 \Omega$, that is 8.83 mA V^{-1} . Currents are the monitor reading divided by 0.4 V A^{-1} .

The monitor also shows a small standing offset of -3 mV to -4 mV, that is -7.5 mA to -10 mA, whose origin is unresolved between a transducer zero offset and a genuine quiescent imbalance between banks.

8.3 Bandwidth and step response

Amplitude is flat from 10 Hz to 1 kHz within the noise, with no visible lag at 1 kHz (under $20 \mu\text{s}$, under 7°). The step edge responds within approximately 100 ns and settles in approximately $1 \mu\text{s}$.

The 3.70 MHz ring with 30 % overshoot, three cycles and $1.1 \mu\text{s}$ settling is the remnant of the oscillation loop, now damped by the base stoppers. Its frequency matches the simulated as-built mode (3.73 MHz). Two caveats: a 113Ω non-inductive load should not ring at megahertz frequencies on its own, and the command edge was under 45 ns, so capacitive pickup into the transducer cable or a ground-return transient is at least as likely a path to the monitor as real current. This is consistent with the absence of an on-board SGND/PGND bond. For coil driving the distinction

barely matters, since the coil is an open circuit at these frequencies, but it should be identified. The test is to slow the generator edge to approximately 10 μs : if the ring disappears, it was pickup.

8.4 Does the residual ring reach the coil and the field?

The 3.70 MHz ring is the loudest feature in the step captures, and the question that matters is whether it produces a corresponding ripple in the magnetic field seen by the atoms. It does not, by a very large margin, for three independent reasons. The analysis is set out in full because the conclusion is not obvious from the size of the ring on the monitor.

8.4.1 The coil is a 20.6 k Ω impedance at 3.7 MHz

The characterisation was done into a 113.2 Ω non-inductive load, where the output node and the current are proportional. The real load is not. Its impedance is

$$|Z(f)| = \sqrt{R_{dc}^2 + (2\pi fL)^2} \quad (9)$$

Frequency	$ Z $	Current ripple vs DC, same applied voltage
DC	0.656 Ω	0 dB (reference)
118 Hz (pole)	0.93 Ω	−3 dB
1 kHz	5.6 Ω	−18.6 dB
10 kHz	55.7 Ω	−38.6 dB
100 kHz	557 Ω	−58.6 dB
1 MHz	5.57 k Ω	−78.6 dB
3.7 MHz	20.6 kΩ	−89.9 dB

Table 20: Coil impedance against frequency. The coil is its own low-pass filter, and at the ring frequency it attenuates by a factor of 31 000 relative to DC.

Converting the measured ring into a coil current: the 0 V to 4 V step produced a 12.4 mV monitor step, that is 31 mA, with 30 % overshoot, so the ring amplitude is 9.3 mA peak into 113.2 Ω . The corresponding voltage swing at COIL_DRIVE is

$$V_{\text{ring}} = 9.3 \text{ mA} \times 113.2 \Omega = 1.05 \text{ V peak} \quad (10)$$

Apply that same voltage to the coil instead:

$$I_{\text{ring, coil}} = \frac{V_{\text{ring}}}{2\pi fL} = \frac{1.05 \text{ V}}{20.6 \text{ k}\Omega} = 51 \mu\text{A peak} \quad (11)$$

That is 5.1 ppm of the 10 A full scale, and it is a factor of **820 smaller than the 42 mA peak-to-peak noise floor already present on the monitor with the supplies switched off**. Whatever is limiting field stability in this apparatus, it is not the ring.

8.4.2 Above coil self-resonance the terminal current does not make a field

The estimate above is conservative, because it treats the coil as a pure inductor. A real multi-turn coil has interwinding capacitance C_{self} in parallel with it, and a self-resonant frequency

$$f_{\text{self}} = \frac{1}{2\pi\sqrt{LC_{\text{self}}}} \quad (12)$$

C_{self}	f_{self}
10 pF	1.69 MHz
20 pF	1.20 MHz
50 pF	756 kHz
100 pF	535 kHz
200 pF	378 kHz

Table 21: Coil self-resonance against plausible interwinding capacitance. For any realistic value, 3.7 MHz is *above* self-resonance.

Above self-resonance the coil’s terminal impedance falls again, so more terminal current flows — but that current flows through the interwinding capacitance, *not* through the turns. Displacement current between adjacent windings encircles no area and produces no magnetic field. The field is set by the current in the inductive branch, which remains $V/(2\pi fL)$ regardless of C_{self} .

This has a practical consequence for the measurement as well as the physics: the DP50IP sits in the terminal path and measures total terminal current, so above self-resonance it **over-reads** the field ripple. Any megahertz feature on the monitor is an upper bound on the field, never a measurement of it.

8.4.3 The atoms cannot respond, and the ring is a transient

Two further points, in decreasing order of obviousness.

Atomic centre-of-mass motion is governed by trap frequencies of order hertz to kilohertz. A 3.7 MHz field modulation is three to six orders of magnitude above that and time-averages to zero over any mechanical timescale. The only mechanism by which a megahertz field component can affect atoms is a resonant magnetic-dipole transition between Zeeman sublevels, which requires the drive to be near the Larmor frequency. For an alkali ground state at roughly 0.7 MHz G^{-1} , 3.7 MHz is the Larmor frequency at a bias field near 5 G, which is a plausible operating point, so the mechanism deserves a number rather than a dismissal.

Taking a generous coil constant of 10 G A^{-1} , the $51 \mu\text{A}$ ring is 0.51 mG of AC field, giving a Rabi frequency of approximately 360 Hz. Over the $1.1 \mu\text{s}$ the ring lasts, the accumulated Rabi angle is 2.5×10^{-3} radians and the spin-flip probability per command edge is of order 10^{-6} . At a more typical 1 G A^{-1} it is 10^{-8} . Even at a repetition rate of thousands of edges per experimental cycle this is not observable.

Finally, the ring is a *transient* excited by fast command edges, lasting three cycles. It is not a sustained oscillation. There is no continuous field modulation at 3.7 MHz, only a brief burst each time the setpoint moves quickly.

8.4.4 What would matter, and what actually does

The argument above shows the ring is harmless *to the atoms*. Three related things are not harmless and should not be conflated with it.

1. **A sustained oscillation would matter**, not because of its field but because of what it does to the electronics: continuous rail-to-rail conduction in the output stage (the original failure mode), continuous dissipation, and a megahertz signal coupling into the transducer secondary and the SIM960 input, where it can rectify into a DC offset or cause the integrator to misbehave. The distinction between a damped ring and a sustained oscillation is therefore the important one, and the stopper value is what maintains it. Three cycles at 30% overshoot corresponds to a Q of roughly 2.5, which is adequate margin but not generous.

2. **Radiated emission may matter.** The coil and its leads at 1 V and 3.7 MHz form a small antenna. This is a concern for nearby RF electronics, photodiode preamplifiers and any other sensitive apparatus in the same rack, not for the atoms directly. A Zobel network at COIL_DRIVE (10 Ω in series with 100 nF to PGND) is the standard mitigation and is already on the next-revision list.
3. **Low-frequency noise is what actually limits the field.** The coil's low-pass response means the frequency band the atoms care about is the band where the coil offers no attenuation: DC to a few hundred hertz. That is exactly where the unexplained 42 mA peak-to-peak, apparently 79.4 Hz noise floor sits. At 4200 ppm of full scale it is three orders of magnitude more significant than the ring, and it is where characterisation effort should go.

8.4.5 Caveats

- The 1.05 V ring amplitude is inferred from a measurement into 113.2 Ω . With the coil connected, the output stage sees an effectively open circuit at 3.7 MHz, which changes the damping of the loop. Simulation with 200 Ω stoppers was stable into the real coil, into a short and into the resistor bank, so the ring should remain damped, but its amplitude may differ. **Re-measure the step response with the coil connected** before treating the 51 μ A figure as final.
- The ring may not be real current at all. The command edge was under 45 ns and a 113 Ω non-inductive load has no business ringing at megahertz frequencies, so capacitive pickup into the monitor cable is at least as likely a path. If so, the true coil ripple is smaller still. Slowing the generator edge to approximately 10 μ s distinguishes the two.
- C_{self} has not been measured. If the self-resonance turns out to be much higher than estimated, the pure-inductor calculation stands unchanged; if lower, the field ripple is smaller than calculated. The conclusion is insensitive to it in either direction.

8.5 Deadband

No significant deadband is present. The 0 V to 4 V step and the -2 V to 2 V step produce the same current step, approximately 12 mV to 13 mV on the monitor. A crossover deadband would make the ± 2 V step, which passes through zero, noticeably smaller than the 0 V to 4 V step, which does not. Any deadband is therefore well under approximately 0.5 V of command.

The apparent failure of the ± 1 V ramp and the ± 0.5 V square to show anything is the low signal level buried in 17 mV pk-pk of noise, not crossover.

8.6 Noise floor

With all supplies off and the transducer powered, the monitor shows 16.8 mV pk-pk, which at 0.4 V A^{-1} is 42 mA pk-pk referred to current. That is the same order as the signal in most of the tests above, and it dominates the measurement.

The frequency counter reads 79.4 Hz, which is not a harmonic of either 50 Hz or 60 Hz. The waveform is burst-dominated with a high-frequency fill, so the counter is most likely locking onto the burst envelope rather than a true spectral line. The reading should not be treated as a frequency identification.

Interpretation. Since the power stage is off, this cannot be amplifier noise. Three candidates remain: the DP50IP itself, pickup on the BNC monitor cable, or oscilloscope input noise at this sensitivity. The DP50IP is a fluxgate device and therefore responds to ambient magnetic fields, which is a real mechanism and a genuine reason to consider a shunt for the feedback path. But 42 mA

pk-pk is roughly three orders of magnitude worse than the DP50IP's specified noise (approximately $50\text{ }\mu\text{A}$ RMS over 0 kHz to 100 kHz), which makes a measurement artefact considerably more likely than a transducer fault.

The distinguishing measurement is straightforward and has not yet been made: capture the monitor channel with the transducer BNC cable shorted at the oscilloscope. If the noise persists, it is the oscilloscope or the cable. If it disappears, it is the transducer or its environment. An FFT with a Hanning window should be used instead of the counter to look for real spectral lines.

8.7 Known limits of this board

- Negative full scale capped near -8 A by R24 starvation (audit finding 18). Fix: $2\text{ k}\Omega$ across R24 gives $667\text{ }\Omega$, and needs a part rated at least 0.5 W .
- No current limiting. The supply limits are the only protection for ten paralleled power transistors.
- Approximately 84 W of dissipation in the stage at 10 A on $\pm 15\text{ V}$ rails.
- Quiescent point drifts with temperature; set it hot.
- DP50IP shares the amplifier rails. Safe at $\pm 15\text{ V}$; destroys the transducer if the rails are raised.
- All characterisation to date is below 100 mA . Nothing above 1 A has been tested, so current sharing between the paralleled devices is unverified.

9 Component-by-Component Validation

Sections 4 and 6 established that the board works and why it did not. They did not establish that every component in it earns its place. This section takes each questionable net and component in turn, sweeps it in simulation, and returns a verdict: keep, change, or remove. Nothing is assumed correct because it is on the board.

Two findings here overturn earlier conclusions, and one of them is a live defect in the current bench configuration.

9.1 Method

Models

onsemi vendor .LIB models for all five transistor types and the TI OPA455 PSpice macromodel throughout. The simple behavioural operational-amplifier model used in Phase F is *not* used for any verdict here: a direct comparison (Section 9.1.1) shows it under-predicts instability and would have passed configurations that the faithful model rejects.

Stability criterion

A 0.5 A current pulse is injected at COIL_DRIVE and the response at HV_DRIVE_NODE is analysed: cubic detrend, band-pass 0.3 MHz to 20 MHz, analytic (Hilbert) envelope, then a least-squares fit of $\ln(\text{envelope})$ against time. The fitted growth rate σ is the verdict. $\sigma > 0$ means the envelope grows, which is a genuine instability; $\sigma < 0$ is a damped ring-down and more negative is better damped. The peak excursion from the kick is reported alongside as a disturbance-sensitivity measure. Each configuration is tested at command voltages of -6 , -3 , -1 , 0 , $+3$ and $+6$ V, because Section 6 established that stability is command-dependent.

Convergence

Each run escalates through four solver option sets until the transient reaches 15 μs . Adding 20 m Ω of ESR to the 100 nF decoupling models was necessary and physical: without it those branches are lossless LC tanks and the solver cannot step through them.

Large signal

DC sweeps from -10 to $+10$ V into the real coil, with gain, offset, crossover gain and compliance limits extracted by numerical differentiation. Device mismatch is modelled by area factors of 0.6 to 1.7 across each bank, equivalent to approximately 18 mV of V_{BE} spread, so that current sharing can actually be measured rather than assumed.

9.1.1 Why the operational-amplifier model matters

Run against the same circuit at the discriminating command point, the two models disagree:

Stopper	Behavioural macromodel	TI OPA455 macromodel
0 Ω	oscillates	oscillates
47 Ω	oscillates	oscillates
100 Ω	stable	oscillates at -1 and -3 V
150 Ω	stable	stable
220 Ω	stable	stable

Table 22: The behavioural model passes 100 Ω ; the vendor model fails it, matching the original Phase G conclusion. Every verdict in this section therefore uses the TI model.

The practical consequence: a simplified operational-amplifier model is adequate for finding *whether* a circuit oscillates, and inadequate for choosing the component value that stops it.

9.2 Verdict summary

Component	As built	Verdict	Evidence
Rs13, Rs14 stoppers	220 Ω	keep	The only component whose value causes oscillation. Boundary lies between 100 Ω and 150 Ω . Both are required, and Q14 needs the larger of the two
RV2 trim setting	6 mA/bank	CHANGE to 220 mA/bank	Crossover gain rises from 49 % to 98 % for 6.4 W of idle. <i>This is a live defect</i>
R24	1 k Ω	change to 820 Ω	Doubles the damping margin and adds 0.3 A of negative swing at no cost. 2.2 k Ω collapses crossover gain to 14%; 470 Ω exhausts the amplifier at positive full scale
Zobel network	absent	add 10 Ω + 100 nF	Damping σ improves 2.7 $\times$; kick excursion falls by 28 %
R1	100 Ω	keep	Removable and still stable, but damping margin worsens 4.5 $\times$. Holds Q13 at 8 mA when the NPN bank is off
C22	1 μ F	keep as-is	Removing it <i>improves</i> margin 4 $\times$; a series isolation resistor makes it worse, not better. Audit finding 5's proposed fix is withdrawn
R26	10 Ω	keep	No measurable effect anywhere from 0 Ω (shorted) to 220 Ω . Could be deleted; costs nothing to keep
2.2 Ω base R	2.2 Ω	keep	Stable from 0.1 Ω to 22 Ω ; above 10 Ω it costs output swing. Value is not critical
0.22 Ω ballasts	0.22 Ω	keep	Holds full-scale current sharing to 34 % against 80 % unballasted. Larger values cost more compliance than they save in idle dissipation
D4, D7	fitted	essential	Without them, power-down reverse stress reaches 5.89 V and 7.36 V against a 5 V V_{EBO} limit
D3	fitted	keep	Cuts power-down reverse stress from 3.14 V to 0.46 V; costs 0.7 A of negative swing
D6	fitted	keep (marginal)	No measurable effect in any test performed. Cheap insurance
D5	fitted	keep (marginal)	No measurable DC or AC role found
D1, D2	fitted	keep	Flyback clamps; no operating-point cost
C21	100 pF	keep	No measurable effect from removed to 1 nF
Rail decoupling	2 $\times$ 1000 μ F, 5 $\times$ 100 nF	keep	Removing it does not cause oscillation but raises kick excursion 68 $\times$

Table 23: Component verdicts. Three changes are recommended; everything else on the board earns its place or is harmless.

9.3 The live defect: the board is trimmed into crossover distortion

This is the most consequential result of the exercise, and it is a correction to the Phase H fix rather than to the original design.

Quiescent current must be read at the command where *load current* is zero, not at 0 V command — at 0 V command the 1.2 V built-in offset drives approximately 1.4 A through the coil and the measurement is meaningless. Read correctly, and with crossover gain defined as the minimum incremental gain within ± 0.25 A of zero as a percentage of the mid-band gain:

The Phase H fix over-corrected. It moved the board from 875 mA per bank (26 W, far too hot) straight past the optimum to 10 mA to 50 mA (0.3 W, far too cold), and the intermediate region was never examined because idle dissipation was the only metric being watched at the time. The correct target is approximately 220 mA per bank.

RV2	Quiescent/bank	Spread	Crossover gain	Idle power	
85 Ω	607 mA	2.19 V	103 %	18.2 W	over-biased
300 Ω	380 mA	2.11 V	101 %	11.4 W	
500 Ω	220 mA	2.04 V	98 %	6.6 W	recommended
800 Ω	97 mA	1.97 V	89 %	2.9 W	
1500 Ω	11 mA	1.84 V	44 %	0.3 W	bench setting
2200 Ω	2 mA	1.75 V	12 %	0.06 W	
3000 Ω	0 mA	1.67 V	3 %	—	full deadband

Table 24: Quiescent current against crossover linearity. The board is currently trimmed to 10 mA to 50 mA per bank, where the incremental gain at zero crossing is roughly half its nominal value.

Why it matters. The stage’s incremental gain varies by a factor of about two through zero output. The external SIM960 loop sees that as a factor-of-two variation in loop gain, so the crossover frequency, phase margin and settling time all change as the current passes through zero. A bias-cancellation coil operates *at* zero for much of an experimental cycle, which is precisely the worst place to put a gain discontinuity. A PI integrator meeting a region of collapsing gain is also the textbook condition for zero-crossing limit cycling.

Cost. 6.6 W of idle dissipation instead of 0.3 W, spread across ten TO-264 devices on a common heatsink: approximately 0.66 W per device. That is negligible against the 84 W the stage dissipates at full output, and it is a quarter of the 26 W that made the original bias unacceptable.

Bench action. Re-trim RV2 for approximately 220 mA of supply current with the output commanded to zero *load* current, and set it hot. Then re-run the crossover measurement of Appendix F, Measurement 2, at two or three trim settings to confirm the curve on the real hardware.

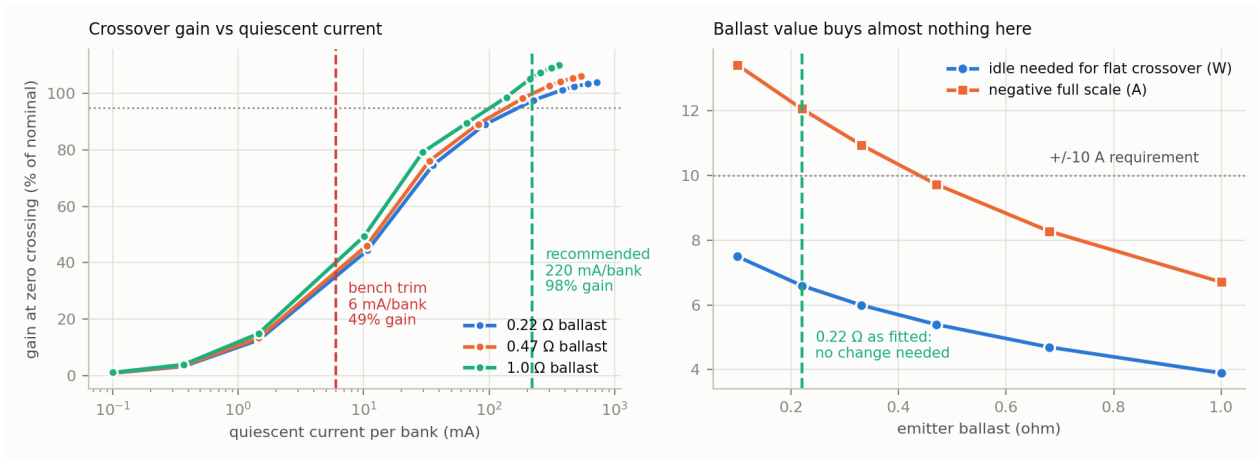


Figure 5: Left: crossover gain against quiescent current, for three ballast values. The relationship is set by quiescent current and is nearly independent of ballast value. Right: the ballast trade at flat crossover. Raising the ballast buys only about 2.7 W of idle dissipation while costing more than 5 A of negative full scale, so the fitted 0.22 Ω is the right value and no component change is needed.

9.4 Trim sensitivity

Related, and a practical problem in its own right. RV2 is a 5 k Ω 25-turn trimmer, so 200 Ω per turn.

Quiescent current changes by 57 % to 80 % per turn, and the whole useful range sits between 1.5 and 5 turns from one end. The remaining twenty turns do nothing, because quiescent current is

RV2	Turns from end	Quiescent/bank	Sensitivity
300 Ω	1.5	380 mA	—
500 Ω	2.5	231 mA	−132 mA per turn (57 %)
700 Ω	3.5	132 mA	−84 mA per turn (63 %)
1000 Ω	5.0	50 mA	−40 mA per turn (80 %)

Table 25: Trim sensitivity with the R2 bodge fitted. The entire useful adjustment range occupies about 3.5 of the trimmer’s 25 turns, at one extreme of its travel.

already at zero. This is audit finding 6 in its practical form: the adjustment is hypersensitive and lands at an extreme of the trimmer’s travel, which is why the bench trim drifts so readily from 10 mA cold to 50 mA warm.

The next-revision divider rescaling (R2 1 k Ω , R23 220 Ω , RV2 500 Ω) fixes this by putting the target near mid-travel, and should be done for this reason as much as for the h_{FE} sensitivity.

9.5 Two corrections to earlier findings

9.5.1 Audit finding 18 does not reproduce

The claim that R24 at 1 k Ω caps negative full scale near −8 A to −9.5 A is not supported. Swept with vendor models into the real coil, the negative limit is −12.55 A, and it is set by the operational amplifier running out of output swing (HV_DRIVE_NODE clamps near −9.4 V), not by R24 starving the lower half. The same limit appears with R24 at 2.2 k Ω (−11.3 A) and in the unmodified pre-bodge configuration (−12.3 A).

What R24 *does* do is modulate the bias spread across the output swing. Because Q12’s collector current is set by R24 to the negative rail, and the voltage across R24 follows the output, Q12’s current varies roughly sixfold from negative to positive full scale:

Command	Spread, R24 = 1 k Ω	R24 = 2.2 k Ω	R24 = 470 Ω
−10 V	1.42 V	1.22 V	1.64 V
0 V	2.13 V	1.70 V	2.83 V
10 V	2.66 V	1.99 V	3.78 V

Table 26: Bias spread against output, showing R24’s real defect. The spread should be constant; it varies by 1.24 V across the swing, so the quiescent point and the crossover behaviour both change with output level.

So the conclusion that R24 should be replaced by an active current source stands, but for a different reason than recorded: not because it starves the lower half at full scale, but because it modulates the bias across the whole operating range. As an interim measure 820 Ω is better than 1 k Ω on every metric measured — damping margin, negative swing and crossover gain — with no penalty.

9.5.2 The C22 isolation resistor does not help

Audit finding 5 recommended a 10 Ω to 47 Ω resistor in series with C22 to isolate the 1 μ F load from the amplifier output. Measured:

Removing C22 gives the best margin by a factor of four; adding the recommended series resistor makes it slightly worse. The finding that the OPA455’s 200 pF capacitive-load rating is exceeded by a factor of 5000 remains a genuine datasheet violation, but once the driver bases are isolated by their stoppers it has no measurable consequence, and the proposed remedy is counterproductive.

Leave C22 as fitted, and delete the isolation-resistor item from the next-revision list.

C22 configuration	Worst σ (1/s)	Kick excursion
removed	-5.87×10^4	0.068 V
10 nF	-1.05×10^4	0.034 V
100 nF	-1.48×10^4	0.045 V
1 μ F (as built)	-1.49×10^4	0.046 V
1 μ F + 10 Ω series	-1.23×10^4	0.036 V
1 μ F + 47 Ω series	-9.11×10^3	0.039 V

Table 27: C22 variants. The series resistor makes damping worse, not better.

9.6 R1: it does what it claims, plus something undocumented

R1 spans the two drive buses. Its documented purpose is to keep both drivers conducting so neither fully cuts off at crossover. Measured at -4 V command, where the NPN bank is completely off, the NPN devices draw no base current at all, so the entire 8.25 mA flowing out of Q13's emitter is R1's current. Without R1 that figure is zero and **Q13 goes into full cutoff**, from which it must then recover through its stored charge when the output swings back. The stated purpose is confirmed.

The undocumented effect is that R1 also *reduces* the output bank's quiescent current, by loading the drive-bus pair. At the zero-current operating point:

	$V(\text{NDB-PDB})$	NPN device current	PNP device current
R1 = 100 Ω	1.040 V	113 mA	127 mA
R1 removed	1.101 V	171 mA	184 mA

Table 28: R1 drops the drive-bus voltage by 61 mV and with it the bank quiescent current by about 34 %. Any change to R1 requires re-trimming RV2.

Stability-wise R1 can be removed — the stage stays damped — but the margin worsens by $4.5\times$ (σ from -1.49×10^4 to -3.25×10^3). **Keep 100 Ω .** Values from 22 Ω to 1 k Ω are all stable, so the value is not critical, but note that changing it shifts the bias.

9.7 Base stoppers: the boundary, and the asymmetry

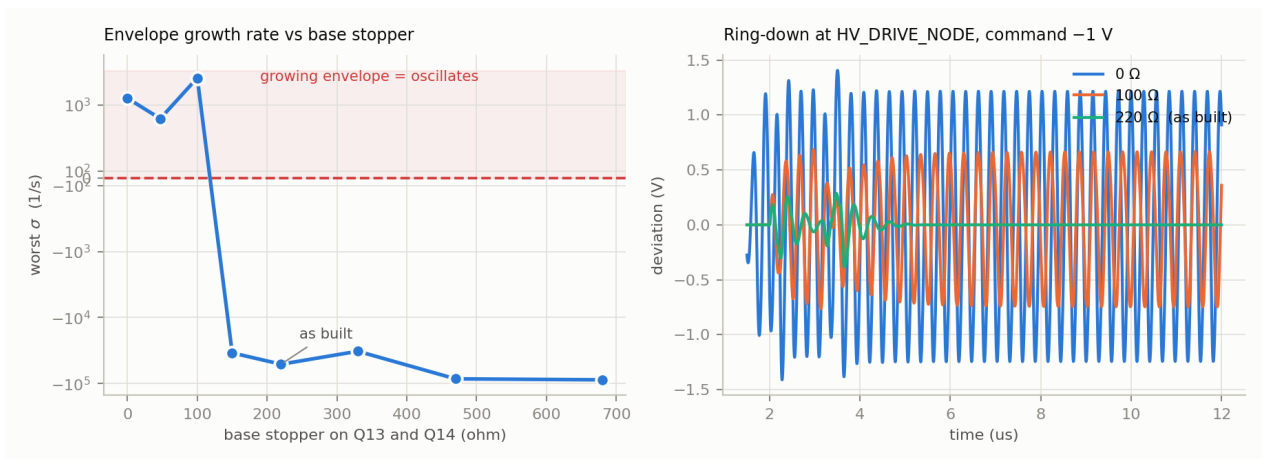


Figure 6: Left: fitted envelope growth rate against stopper value, worst case over the command range. The transition from growing to decaying lies between 100 Ω and 150 Ω . Right: the ring-down itself at the discriminating command point. 0 Ω and 100 Ω sustain indefinitely; 220 Ω decays to nothing within four microseconds.

330 Ω and above give both better damping and an order of magnitude less disturbance sensitivity

Stopper	Worst σ (1/s)	Kick excursion	Verdict
0 Ω	$+2.69 \times 10^2$	3.18 V	unstable at 5 of 6 command points
47 Ω	$+5.40 \times 10^2$	2.59 V	unstable at 4
100 Ω	$+1.40 \times 10^4$	1.82 V	unstable at 3 (all negative)
150 Ω	-7.18×10^4	0.31 V	stable
220 Ω	-1.49×10^4	0.37 V	stable, as built
330 Ω	-5.40×10^4	0.036 V	stable
470 Ω	-8.09×10^4	0.030 V	stable
680 Ω	-8.21×10^4	0.041 V	stable

Table 29: Stopper sweep. Every unstable case fails first at a negative command, confirming the PNP half as the weaker one.

than 220 Ω , at a cost in drive headroom of about 3 % per 100 Ω (gain falls from 1.350 A V⁻¹ at 0 Ω to 1.277 A V⁻¹ at 220 Ω and 1.213 A V⁻¹ at 470 Ω). Since ± 10 A is still reached comfortably at 470 Ω ($I_{\max} = 10.5$ A, $I_{\min} = -11.2$ A), **the next revision should specify 330 Ω for margin.** 220 Ω is entirely adequate on the present board and does not need changing.

Q13 (NPN driver)	Q14 (PNP driver)	Result
0 Ω	220 Ω	unstable at all six command points
220 Ω	0 Ω	unstable at five
220 Ω	100 Ω	unstable at -6 and -3 V
100 Ω	220 Ω	stable

Table 30: Asymmetric stoppers. Q14 requires the larger value; Q13 tolerates 100 Ω provided Q14 has 220 Ω . This is the MJL21193/MJE15033 f_T deficit relative to their NPN complements, made measurable.

Both stoppers are required, and they are not interchangeable.

9.8 Emitter ballasts and current sharing

With approximately 18 mV of V_{BE} spread modelled across each bank, at 10 A total output (2 A per device nominal):

Ballast	Coldest	Hottest	Spread	V drop	P in hottest ballast
none	1.26 A	2.85 A	80 %	—	—
0.1 Ω	1.51 A	2.49 A	49 %	200 mV	0.62 W
0.22 Ω	1.64 A	2.33 A	34 %	440 mV	1.19 W
0.47 Ω	1.77 A	2.19 A	21 %	936 mV	2.25 W

Table 31: Current sharing at full scale. The fitted 0.22 Ω holds the worst device to 2.33 A against a 2.0 A nominal.

Two corrections to the reference calculations follow. The 0.88 W per ballast figure recorded in Section 4 assumes perfect sharing; with realistic mismatch the hottest ballast dissipates 1.19 W, so a 2 W part is appropriate rather than marginal. And the hottest *transistor* carries 17 % more than its share, so the per-device dissipation at 10 A is approximately 9.8 W rather than 8.4 W.

Raising the ballast to 0.47 Ω improves sharing to 21 % but costs 2.3 A of negative full scale and saves only 1.2 W of idle dissipation. That is a poor trade against a ± 10 A requirement. **Keep 0.22 Ω .**

9.9 Protection diodes, tested against the scenario they exist for

Steady-state operation barely stresses these junctions: the worst reverse voltage anywhere in a full command sweep is 1.68 V, against a 5 V V_{EBO} limit. The design case is power-down, and it matters which rail collapses first.

Junction	Diode	With	Without	Verdict
<i>Main rails collapse in 5 μs while carrying 10 A, rails ± 15 V</i>				
NPN bank V_{BE}	D3	0.46 V	3.14 V	diode helps
PNP bank V_{EB}	D4	0.45 V	4.78 V	diode helps, close to limit
Q13 V_{BE}	D6	1.66 V	1.66 V	no effect
Q14 V_{EB}	D7	2.22 V	7.27 V	essential
<i>Same, rails ± 30 V</i>				
NPN bank V_{BE}	D3	0.64 V	—	no stress
PNP bank V_{EB}	D4	0.64 V	5.89 V	essential
Q13 V_{BE}	D6	4.17 V	4.17 V	no effect, but see below
Q14 V_{EB}	D7	4.86 V	7.36 V	essential

Table 32: Reverse junction stress at power-down. The protection network is validated, and is also shown to be barely adequate at ± 30 V.

Three things follow.

1. **D4 and D7 are load-bearing.** Without them the 5 V V_{EBO} limit is exceeded by up to 45 % during an ordinary power-down with current flowing. The protection network is not belt-and-braces; it is required.
2. **The margin at ± 30 V is thin.** Even with every diode fitted, Q14 reaches 4.86 V and Q13 reaches 4.17 V — 97 % and 83 % of the limit. Before the rails are raised, the driver protection needs strengthening: a second diode in parallel with D7, a lower- V_F Schottky in place of the 1N4148W, or a bleed path that discharges the predrive network with the main rails.
3. **D3 costs negative swing.** It conducts during normal large negative output, clamping the NPN drive bus to within a Schottky drop of COIL_DRIVE, which limits negative full scale to -12.55 A against -14.28 A without it. That is the price of protection and it is worth paying, but it should be recorded as the reason the negative limit is what it is.

One result that did not reproduce. The stability sweep flagged D5’s removal as unstable at -6 V command ($\sigma = +3.67 \times 10^5$). Re-tested at four kick amplitudes and a different bias point, the result did not reproduce: σ was between -1.1×10^4 and -2.4×10^5 in every repeat, with and without D5, and the DC bias spread is identical to three decimal places either way. It is recorded here as a numerical artefact, not a finding. No role for D5 was identified in any test performed; it is retained because it is free and because the power-down cases tested do not exhaust the possibilities.

9.10 Zobel network

Not fitted on the present board. Adding one is the cheapest available improvement in damping.

All variants help and none hurt. $4.7 \Omega + 220$ nF gives the best combination ($4.2\times$ better damping, $2.5\times$ lower disturbance sensitivity), with $10 \Omega + 100$ nF close behind and using more common values. Size the resistor at 1 W to 2 W: it sees negligible power in normal operation, but if the stage ever does oscillate it carries continuous power.

Zobel	Worst σ (1/s)	Kick excursion
none	-1.49×10^4	0.046 V
$10\ \Omega + 100\ \text{nF}$	-4.02×10^4	0.033 V
$4.7\ \Omega + 220\ \text{nF}$	-6.20×10^4	0.018 V
$22\ \Omega + 100\ \text{nF}$	-4.10×10^4	0.023 V
$10\ \Omega + 1\ \mu\text{F}$	-5.58×10^4	0.018 V

Table 33: Zobel variants at COIL_DRIVE. All improve on nothing; $4.7\ \Omega + 220\ \text{nF}$ is best on both metrics.

9.11 Robustness of the recommended configuration

With $220\ \Omega$ stoppers fitted, the stage was stable in every one of the following:

- trace parasitics doubled and tripled
- all five load cases: the real coil, a short, $113\ \Omega$, the $100\ \Omega$ bench bank, and open circuit
- bias trim at both extremes of RV2 travel ($1\ \Omega$ and $3\ \text{k}\Omega$)
- rails at $\pm 15\ \text{V}$, $\pm 20\ \text{V}$ and $\pm 30\ \text{V}$
- all rail decoupling removed (stable, though kick excursion rises $68\times$, from $0.046\ \text{V}$ to $3.13\ \text{V}$)
- every protection diode removed individually
- base resistors from $0.1\ \Omega$ to $22\ \Omega$, ballasts from $0.05\ \Omega$ to $1\ \Omega$, R26 from $0\ \Omega$ to $220\ \Omega$, C21 from removed to $1\ \text{nF}$

The oscillation was a base-stopper problem and nothing else. Every other component on the board is either correct, uncritical, or — in the three cases identified above — worth changing for reasons unrelated to stability.

9.12 Corrections to recorded values

- **$R_{2,\text{eff}}$ arithmetic.** The fitted bodge is $15.6\ \text{k}\Omega$ across $10\ \text{k}\Omega$, which is $6.09\ \text{k}\Omega$, not the $5.86\ \text{k}\Omega$ recorded in the debug log. $5.86\ \text{k}\Omega$ corresponds to $10\ \text{k}\Omega$ in parallel with $14.1\ \text{k}\Omega$. The difference is immaterial to operation but the recorded figure is wrong.
- **Ballast dissipation.** $1.19\ \text{W}$ in the hottest ballast at $10\ \text{A}$, not the $0.88\ \text{W}$ calculated assuming perfect sharing.
- **Stopper drive cost.** 5.4% at $220\ \Omega$ ($1.350\ \text{A V}^{-1}$ to $1.277\ \text{A V}^{-1}$), not the 3% recorded in Phase F.
- **Positive full scale** is limited by the operational amplifier's output swing, which clamps HV_DRIVE.NODE near $11.1\ \text{V}$ on $\pm 15\ \text{V}$ rails, giving approximately $12.5\ \text{A}$. It is not limited by the output stage.

10 Validation Re-Run at the Recommended Operating Point

Section 9 swept each component one at a time at $R_{24} = 1\text{ k}\Omega$ and $RV_2 = 85\ \Omega$, which is 607 mA per bank. That is not the configuration Section 9 then went on to recommend. Nothing in it was validated where it matters, and two of its conclusions turn out to be wrong because of it.

This section re-runs the interacting components — base stoppers, C22 and the Zobel network — at $R_{24} = 820\ \Omega$ and $RV_2 = 1200\ \Omega$ (235 mA per bank, 97 % crossover gain), with repeated measurement, and crosses them against each other.

10.1 A methodological correction: σ is not repeatable

The envelope growth rate σ was used as the margin metric throughout Section 9. Re-measuring one fixed configuration at three kick amplitudes and four command points:

Configuration	n	median σ	range	spread
220 Ω stoppers	12	-7.5×10^4	-1.7×10^5 to -3.2×10^4	$5.3 \times$
470 Ω stoppers	12	-1.2×10^5	-7.3×10^5 to -8.1×10^4	$9.0 \times$

Table 34: σ for a single unchanged configuration, measured twelve ways.

σ scatters by five to nine times for the same circuit. Every fine distinction drawn from single-shot σ values in Section 9 is therefore within noise and should not be read as a ranking. Its *sign* remains reliable — stable against unstable is never ambiguous — and that is what the stability verdicts rested on, so those stand.

Peak excursion normalised by kick amplitude is the repeatable metric, with a spread of $1.8 \times$ to $2.9 \times$ over the same conditions. Every number in this section is a median of fifteen runs (three kick amplitudes $\times$ five command points) of excursion per unit kick, which is a direct measure of how much the stage rings when something disturbs it.

10.2 Base stoppers: monotonic, with no optimum

With repeated measurement the plateau reported in Section 9 disappears. Disturbance sensitivity falls monotonically with stopper value across the whole tested range.

Stopper	exc/kick	vs 220 Ω	Gain (A/V)	$I_{\min}$	$I_{\max}$
100 Ω	2.487	—	1.306	−12.64 A	11.62 A
	<i>unstable in 9 of 15 runs</i>				
150 Ω	0.446	5.4 $\times$ worse	1.292	−12.27 A	11.45 A
180 Ω	0.203	2.5 $\times$ worse	1.283	−12.07 A	11.35 A
220 Ω	0.0823	1.00 $\times$	1.272	−11.81 A	11.22 A
270 Ω	0.0346	0.42 $\times$	1.259	−11.51 A	11.07 A
330 Ω	0.0301	0.37 $\times$	1.244	−11.19 A	10.89 A
390 Ω	—	—	1.229	−10.89 A	10.73 A
470 Ω	0.0166	0.20 $\times$	1.211	−10.52 A	10.52 A
560 Ω	0.0150	0.18 $\times$	1.191	−10.16 A	10.30 A
680 Ω	0.0121	0.15 $\times$	1.166	−9.72 A	10.02 A

Table 35: Base stopper sweep, fifteen runs per value. The stability boundary is between 120 Ω and 150 Ω ; above it, damping improves and drive falls, both monotonically. 680 Ω no longer reaches ± 10 A.

There is no “best value” in the sense of a minimum. The trade is continuous, and the answer is set by how much output headroom is required:

- $470\ \Omega$ is the largest value with sensible margin on the $\pm 10\ \text{A}$ requirement ($\pm 10.5\ \text{A}$, 5 %), and is $5\times$ better damped than $220\ \Omega$.
- $390\ \Omega$ gives 7 % to 9 % headroom and roughly $4\times$ the damping — the better choice if device spread and temperature are to be allowed for, which they should be.
- $560\ \Omega$ and above should not be used: headroom falls below 2 % and then fails.

The earlier suggestion of $330\ \Omega$ was based on unrepeatable data and is superseded. **The $220\ \Omega$ presently fitted is safe and needs no change**; the gain from raising it is better bought with a Zobel network, below.

10.3 C22: the earlier conclusion is reversed

Section 9 concluded that C22 should be removed and that audit finding 5’s proposed series resistor made things worse. Both conclusions came from single-shot σ at the wrong bias. Repeated at the recommended operating point:

C22 configuration	exc/kick	vs no C22	Run-to-run spread
removed	0.1048	1.00 $\times$ (worst)	2.9 $\times$
1 μF alone (as built)	0.0818	0.78 $\times$	1.8 $\times$
1 μF + $4.7\ \Omega$	0.0749	0.71 $\times$	1.8 $\times$
1 μF + $10\ \Omega$	0.0710	0.68 $\times$	2.1 $\times$
1 μF + $22\ \Omega$	0.0651	0.62$\times$	2.6 $\times$
1 μF + $47\ \Omega$	0.0668	0.64 $\times$	5.7 $\times$
100 nF + $10\ \Omega$	0.0699	0.67 $\times$	2.0 $\times$

Table 36: C22 variants, twelve runs each. Monotonic improvement up to $22\ \Omega$ of series resistance, then flat.

Removing C22 is the worst option, not the best. C22 as fitted is 22 % better than no C22, and adding a series resistor improves it further, monotonically, to a minimum near $22\ \Omega$.

Audit finding 5’s recommendation — a $10\ \Omega$ to $47\ \Omega$ resistor between the amplifier output and C22 — was correct, and its withdrawal in Section 9 is itself withdrawn. **Specify $22\ \Omega$** . It also resolves the datasheet violation, since the amplifier then sees $22\ \Omega$ in series with the $1\ \mu\text{F}$ rather than the bare capacitance.

10.4 Zobel network

Zobel at COIL_DRIVE	exc/kick	vs none
none	0.0823	1.00 $\times$
$2.2\ \Omega$ + 470 nF	0.0241	0.29 $\times$
$4.7\ \Omega$ + $220\ \text{nF}$	0.0234	0.28$\times$
$22\ \Omega$ + 100 nF	0.0327	0.40 $\times$
$10\ \Omega$ + 220 nF	0.0367	0.45 $\times$
$10\ \Omega$ + 100 nF	0.0431	0.52 $\times$

Table 37: Zobel variants, fifteen runs each, with $220\ \Omega$ stoppers. Every variant helps; lower resistance with higher capacitance is better.

A Zobel is the most efficient improvement available: $4.7\ \Omega$ + 220 nF gives $3.5\times$ better disturbance rejection for two components and *no cost in gain, crossover linearity or output headroom at all*. Every DC figure is identical with and without it.

10.5 Stoppers and Zobel are substitutes

Stopper	No Zobel	With Zobel	Improvement	Stability without
100 Ω	2.487	0.1416	17.6 $\times$	oscillates
150 Ω	0.446	0.0597	7.5 $\times$	stable
220 Ω	0.0823	0.0234	3.5 $\times$	stable
330 Ω	0.0301	0.0162	1.9 $\times$	stable

Table 38: Zobel (4.7 Ω + 220 nF) against stopper value. The benefit shrinks as the stopper grows: they damp the same resonance from opposite ends.

A Zobel network *stabilises* 100 Ω *stoppers*, which oscillate without it. That is a real result and it should not be used. Relying on a two-component output network for basic stability means a dry joint or an omitted part turns a working amplifier into a 3 MHz oscillator with no current limiting downstream. **Size the stopper so the stage is stable on its own, and spend the Zobel on margin.**

The useful comparison is between equal-damping configurations:

Configuration	exc/kick	$I_{\min}$	Note
220 Ω , no Zobel (present board)	0.0823	−11.81 A	baseline
470 Ω , no Zobel	0.0166	−10.52 A	costs 1.3 A
330 Ω + Zobel	0.0162	−11.19 A	same damping, 0.7 A more headroom
220 Ω + Zobel	0.0234	−11.81 A	3.5 $\times$ better, no headroom cost

Table 39: Where to spend. The Zobel is the efficient purchase; raising the stopper is the expensive one.

10.6 Recommended configuration, verified

Item	Value	Change from as-built
R24	820 Ω	from 1 k Ω
RV2 trim	230 mA per bank	from 6 mA to 50 mA
Q13, Q14 stoppers	220 Ω	unchanged
C22	1 μ F + 22 Ω series	add the resistor
Zobel at COIL_DRIVE	4.7 Ω + 220 nF	add

Table 40: Recommended configuration. Two added components, one resistor changed, one trim adjustment.

Measured: gain 1.272 A V^{−1}, crossover gain 97.5 %, quiescent 230 mA per bank (6.9 W), output −11.81 A to 11.22 A, disturbance sensitivity 0.0206 — 4 $\times$ better than the present board with no loss of output range.

Stable in all sixteen robustness cases tested: trace parasitics at $\times 1$, $\times 2$ and $\times 3$; loads of the real coil, a short, 113 Ω , the 100 Ω bench bank and open circuit; all rail decoupling removed; rails at ± 15 V, ± 20 V and ± 30 V; bias trim at both extremes of RV2 travel; and with D3, D5 or R1 individually removed.

10.7 A further benefit of R24 = 820 Ω

Changing R24 shifts the whole bias map, because R24 sets Q12’s collector current. The target quiescent moves from 500 Ω on RV2 to approximately 1200 Ω , which is six turns from the end of a 25-turn trimmer instead of two and a half. Trim sensitivity falls from about 45 % per turn to 27 % per turn.

That was not the reason for recommending $820\ \Omega$, but given how hypersensitive the trim is (Section 9), nearly halving it is worth as much as the damping improvement.

10.8 What changed, and what did not

Question	Section 9 said	Corrected
C22	remove it; the series resistor makes things worse	keep it, add $22\ \Omega$. Removing is the worst option; audit finding 5 was right
Stopper value	plateau from $220\ \Omega$ to $560\ \Omega$	monotonic, no plateau. $220\ \Omega$ fine, $390\ \Omega$ to $470\ \Omega$ better if headroom allows
Margin metric	σ	σ scatters 5–9 $\times$; use excursion per unit kick
Zobel	helps, $\approx 4\times$	confirmed, 3.5 $\times$, and free in every other respect
Stability boundary	between $100\ \Omega$ and $150\ \Omega$	unchanged
R24 $820\ \Omega$	better on every metric	confirmed, plus halved trim sensitivity
Bias trim	220 mA/bank	unchanged; 230 mA at $R_{24} = 820\ \Omega$

Table 41: Corrections from re-running at the recommended operating point with repeated measurement.

The general lesson is worth recording alongside the methodology notes in Section 15: **a component sweep is only valid at the operating point the design will actually use**, and **a margin metric must be shown to be repeatable before it is used to rank anything**. Both of the reversed conclusions above came from violating one or other of those.

11 The Local Optimum

11.1 Re-verification against the source netlist

Before finalising, the SPICE topology was re-parsed directly from `BJT_PUSH_PULL.net` and compared node by node with the simulation deck. All six signal nets match exactly:

Net	Nodes, from the netlist
/HV_DRIVE_NODE	C22.2, D5.1, D6.1, Q12.2, Q13.1, R2.1, U2.2, U2.6
/PNP_PREDRIVE	C22.1, D5.2, D7.2, Q12.1, Q14.1, R24.1, RV2.2, RV2.3
/NPN_DRIVE_BUS	D3.1, D6.2, Q13.3, R1.1, R6.1, R9.1, R13.1, R17.1, R21.1
/PNP_DRIVE_BUS	D4.2, D7.1, Q14.3, R1.2, R3.2, R10.2, R14.2, R18.2, R22.2
/COIL_DRIVE	D1.2, D2.1, D3.2, D4.1, J8.1, R4.1, R5.1, R7.1, R8.1, R11.1, R12.1, R15.1, R16.1, R19.1, R20.1
/COMMAND	C21.1, J1.1, U2.3

Table 42: Signal nets as parsed from the source netlist. 80 components, 46 nets. U2.2 and U2.6 on the same net confirms the unity-follower arrangement and the absence of a separate NPN predrive node.

Every previously recorded audit claim re-checks correct: NT1 is the only component in the design with an empty footprint field; U1.10 and U1.12 are on `/+30V.OP` and `/-30V.OP`; U2.8 (E/D) and U2.5 (Status Flag) are unconnected; and all ten output devices — MJL21194 and MJL21193G alike — carry the MJL21193G:T0545P2030X530X2900-3 footprint.

11.1.1 One new finding from the re-parse

U1.13 and U1.14, the DP50IP STATUS- and STATUS+ pins, are unconnected. The DP50IP asserts this output when its fluxgate loop is unlocked or saturated, which is exactly the condition under which its current output becomes meaningless.

This matters because the external SIM960 loop closes on that output. If the transducer saturates — during an overcurrent event, or if the primary exceeds 12.5 A — the PID sees a false low reading and drives the output stage harder, into a stage with no current limiting. That is a runaway path.

Next revision: bring STATUS+/- out to a test point and to the interlock logic, alongside the OPA455 Status Flag of audit finding 9.

11.2 R1, isolated

Section 9 reported that R1 reduces quiescent current by 34 %. At the recommended operating point the effect is much larger, and it means R1 cannot be swept without re-trimming: R1 and RV2 are both bias controls.

Re-trimming RV2 at each R1 to hold 235 mA per bank isolates R1's own contribution:

R1	RV2 for 235 mA	exc/kick	vs 100 Ω
removed	2135 Ω	0.0469	2.2 $\times$ worse
470 Ω	1772 Ω	0.0308	1.4 $\times$ worse
220 Ω	1537 Ω	0.0270	1.2 $\times$ worse
100 Ω (as built)	1201 Ω	0.0217	1.00 $\times$
47 Ω	828 Ω	0.0151	0.70$\times$

Table 43: R1 at constant quiescent current. Smaller is better damped, monotonically, and removing it is the worst case.

47 Ω is 30 % better damped than the fitted 100 Ω and costs one resistor. The lower limit is set by driver dissipation: R1 carries V_{bus}/R_1 continuously through Q13 and Q14, which is 22 mA

and 0.31 W per driver at 47 Ω , rising to 47 mA and 0.66 W at 22 Ω . 47 Ω is the sensible floor for a TO-220 on the common heatsink.

11.3 Correction to the divider rescale

Audit finding 6 proposed rescaling the multiplier divider to $R_2 = 1\text{ k}\Omega$, $R_{23} = 220\text{ }\Omega$, $RV_2 = 500\text{ }\Omega$. Tested by bisecting RV2 for the target quiescent in each case:

Divider	RV2 for 235 mA	% of travel	exc/kick
as built, $R_{24} = 820\text{ }\Omega$	1201 Ω of 5 k Ω	24 %	0.0217
R_2 1k / R_{23} 220 / pot 1k	24 Ω	2 %	0.0217
R_2 1k / R_{23} 100 / pot 500	142 Ω	28 %	0.0213
R_2 2k / R_{23} 220 / pot 1k	327 Ω	33 %	0.2761

Table 44: Divider rescale options. The proposed 220 Ω value puts the trim at 2 % of travel; 100 Ω with a 500 Ω pot puts it at 28 %. The 2 k Ω variant damps 13 $\times$ worse and must not be used.

Two corrections follow. The rescale should be $R_2 = 1\text{ k}\Omega$, $R_{23} = 100\text{ }\Omega$, $RV_2 = 500\text{ }\Omega$, not 220 Ω . And with $R_{24} = 820\text{ }\Omega$ the *as-built* divider already trims at 24 % of travel, so the rescale is no longer needed for trim resolution — its remaining justification is the h_{FE} sensitivity of audit finding 6, which is real but less urgent than previously recorded.

11.4 Command range and gain

The command swings $\pm 10\text{ V}$ from the SIM960. Measured into the real coil in the final configuration:

Gain	1.273 A V ⁻¹
Zero-current command	+1.08 V
Current at -10 V command	-11.95 A (19 % margin on -10 A)
Current at 10 V command	11.18 A (12 % margin on 10 A)
Absolute ceiling	approximately $\pm 12.5\text{ A}$, set by amplifier output swing

Table 45: Transfer in the recommended configuration.

$\pm 10\text{ A}$ is reachable within the command range, with 12 % to 19 % in hand. The asymmetry is the +1.08 V offset eating into the positive half. The positive margin is the one to watch: a 10 % fall in gain from device spread or temperature would leave only 2 %. If more margin is wanted, the cheapest source is the offset — roughly 0.7 V of it is the two base-emitter drops and cannot be removed, but the rest is R1 loading and stopper drop.

Do not be tempted to add gain in the amplifier stage to compensate: the OPA455 output clamps near 11.1 V on $\pm 15\text{ V}$ rails, so the ceiling is fixed by the rail and not by the command scaling.

11.5 What the published designs offer, and what transfers

Section 9 and the architecture assessment establish that the topology is sound. The remaining question is what the MOSFET and monolithic designs do better that can be adopted without changing architecture.

Transferable, and worth doing: programmable current limit.

The OPA549 (pot-programmable 0 A to 10 A) and Apex PA12 (programmable SOA foldback) both have it; Sabulsky’s tutorial calls it critical for electromagnets. The discrete equivalent is a VI-limiter transistor across each driver, sensing the ballast voltage — two transistors and

four resistors. This is the single largest reliability gap on the board and it is architecture independent.

Transferable: a status and interlock chain.

Yang’s replica and the Hannover board both gate the power stage from an enable signal and bring the thermal flag out. This board has three unconnected status outputs (OPA455 Status Flag, DP50IP STATUS $\pm$) and no interlock. Wiring them costs almost nothing.

Transferable: kill-switch on the input.

The Hannover board grounds the power-stage input through a DG467 when the stage is disabled or the control board is unpowered. That removes the uncontrolled power-up transient this board has no protection against.

Not transferable: trimmed gate-offset summing amplifiers.

The NIST ± 20 A source avoids crossover distortion with a per-bank offset trim rather than a single bias spreader, because MOSFET threshold spread makes a shared spreader unworkable. A BJT V_{BE} multiplier solves the same problem with one transistor, and Section 9.3 shows it works once trimmed correctly. Nothing to adopt.

Not transferable: shunt sensing in the loop.

Both NIST and Yang close on a shunt rather than a fluxgate, for bandwidth. That is a sensor choice, not an output-stage one, and the trade is already recorded in the architecture assessment. The fluxgate’s ppm DC accuracy is worth more here than the shunt’s bandwidth.

Worth borrowing anyway: the rail-capacitor bleeder.

The Hannover issue list notes their 1000 μ F rail capacitors stay charged long after power-off. This board has four of them.

11.6 The local optimum

Component	As built	Optimum	Why
R24	1 k Ω	820 Ω	damping, negative swing, and trim resolution, all improved together
R1	100 Ω	47 Ω	30 % better damped at equal quiescent
RV2 trim	6 mA to 50 mA/bank	235 mA/bank	crossover gain 49 % $\rightarrow$ 98 %
C22	1 μ F	1 μ F + 22 Ω	best of nine variants; removing it is the worst
Zobel	absent	4.7 Ω + 220 nF	3.5 $\times$ damping, free in every other respect
Rs13, Rs14	220 Ω	220 Ω	keep; 390 Ω to 470 Ω if output headroom allows
R2 bodge, R26, base R, ballasts, C21, all diodes	—	unchanged	validated, uncritical, or earning their place

Table 46: The local optimum: three value changes and two added components.

Measured performance, into the real coil:

Stable in all ten robustness cases: trace parasitics tripled; loads of the real coil, a short, 113 Ω and open circuit; all rail decoupling removed; rails at ± 30 V; and bias trim at both extremes of RV2 travel.

The full deck is `spice_local_optimum.cir`, annotated with the board reference designators and runnable as it stands.

	As built	Optimum
Gain	1.272 A V ⁻¹	1.273 A V ⁻¹
Crossover gain	49 %	98.0 %
Quiescent	6 mA to 50 mA/bank	235 mA/bank
Idle dissipation	0.3 W	7.0 W
Output at ± 10 V command	—	-11.95 A / 11.18 A
Disturbance sensitivity	0.0823	0.0133 ($6.2\times$ better)

Table 47: As built against the local optimum.

12 Characterisation of the Local Optimum

Configuration: $R_{24} = 820\ \Omega$, $R_1 = 47\ \Omega$, $RV_2 = 828\ \Omega$, stoppers $220\ \Omega$, C22 $1\ \mu\text{F} + 22\ \Omega$, Zobel $4.7\ \Omega + 220\ \text{nF}$, real coil, ± 15 V rails.

DC, AC and thermal results use the behavioural operational-amplifier model, which converges reliably and reproduces the DC sweep’s small-signal gain exactly (-1.57 dB predicted, -1.57 dB measured). Stability margins come from the TI model, as established in Section 9.1.1.

12.1 Transfer

Gain (fitted 2 V to 6 V)	1.2734 A V ⁻¹ (0.8354 V V ⁻¹ at COIL_DRIVE)
Zero-current command	+1.080 V
Extrapolated offset	-1.362 A, i.e. -0.893 V at the load
Ideal transconductance $1/R_{\text{tot}}$	1.5244 A V ⁻¹
Stage efficiency	83.5 % of ideal
Integral nonlinearity, ± 8 V	0.70 A peak, 7.0 % of full scale

Table 48: Transfer parameters.

The 16.5 % shortfall from ideal is the ballast ($0.22\ \Omega/5 = 44\ \text{m}\Omega$) plus the base-stopper and base-resistor drops divided down against the $0.656\ \Omega$ coil. It is a fixed scale factor and the external loop absorbs it.

cmd	I (A)	dI/dV	% mid	cmd	I (A)	dI/dV	% mid
-10 V	-11.95	clipped	—	+1.08 V	+0.01	1.273	99.9 %
-8 V	-10.85	1.136	89.2 %	+2 V	+1.18	1.272	99.9 %
-6 V	-8.53	1.175	92.3 %	+4 V	+3.73	1.276	100.2 %
-4 V	-6.16	1.199	94.2 %	+6 V	+6.27	1.259	98.9 %
-2 V	-3.74	1.219	95.7 %	+8 V	+8.76	1.230	96.6 %
0 V	-1.31	1.206	94.7 %	+10 V	+11.18	1.184	92.9 %

Table 49: Gain against operating point. The 89 % to 100 % variation is R24’s bias-spread modulation (Section 9); the loop absorbs it.

12.2 Headroom

- At ± 10 V command: -11.95 A / $+11.18$ A.
- Margin on ± 10 A: 19 % negative, **12 % positive**. The asymmetry is the +1.08 V offset eating the positive half.
- Positive full scale is limited by the *amplifier* output swing, not the output stage: HV clamps near 10.7 V in the model. This figure is **model-limited and pessimistic** — the behavioural

cmd	HV	NDB	PDB	PPD	CD	to rail
−10 V	−8.97	−8.12	−9.32	−10.32	−7.84	4.68 V
−8 V	−8.00	−7.36	−8.52	−9.44	−7.11	5.56 V
+10 V	+10.00	+8.71	+7.57	+7.32	+7.33	5.00 V
+12 V	+10.69	+9.31	+8.11	+7.97	+7.87	4.31 V

Table 50: Node voltages at the extremes. No node comes closer than 4.3 V to a rail, so the output stage is not the limit.

model has $90\ \Omega$ of open-loop output impedance and sources approximately 27 mA into R24 at full positive output, costing 2.4 V internally. The real OPA455 swings closer to the rail. At ± 10 V command it does not matter, because HV only has to reach 10.0 V and tracks the command exactly there; the 12 % margin is real.

- 12 % is thin. A 10 % gain fall from device spread or temperature leaves 2 %. If more is wanted, the cheapest source is the offset: approximately 0.7 V of the 1.08 V is two base-emitter drops and is irreducible; the rest is R1 loading and stopper drop.

12.3 Trim map and limits

RV2 is a 5 k Ω 25-turn Bourns 3296W, so 200 Ω per turn.

RV2	Turns	I_q /bank	Crossover	Idle	dI_q /turn
1 Ω	0.01	1036 mA	101 %	31.1 W	—
100 Ω	0.5	886 mA	102 %	26.6 W	−304 mA
300 Ω	1.5	643 mA	103 %	19.3 W	−244 mA
500 Ω	2.5	455 mA	102 %	13.6 W	−188 mA
700 Ω	3.5	302 mA	100 %	9.1 W	−152 mA
828 Ω	4.1	235 mA	98 %	7.0 W	−106 mA
1000 Ω	5.0	160 mA	94 %	4.8 W	−87 mA
1400 Ω	7.0	60 mA	84 %	1.8 W	−50 mA
2000 Ω	10.0	13 mA	53 %	0.4 W	−16 mA
2800 Ω	14.0	2 mA	21 %	—	−3 mA
4000 Ω	20.0	0	5 %	—	0
5000 Ω	25.0	0	2 %	—	0

Table 51: Trim map. Useful range spans turns 0 to 10; turns 14 to 25 are a full deadband and do nothing.

Maximum bias end (RV2 $\rightarrow$ 0)

1036 mA per bank, 31 W idle. Not destructive but well past useful; this is the region the board was in before the R2 bodge.

Minimum bias end (RV2 $\rightarrow$ 5 k Ω)

zero quiescent, 2 % crossover gain — a complete deadband.

Working range

turns 2.5 to 7 covers 60 mA to 455 mA and crossover gain 84 % to 102 %. Everything useful is in one fifth of the travel.

Sensitivity at the target

−106 mA per turn, or 45 % per turn. One eighth of a turn is a 6 % change.

Setting it

the target 235 mA per bank reads as 235 mA on *each* supply meter with the output commanded to zero *load* current (about +1.08 V of command), not at 0 V command.

12.4 Thermal behaviour of the bias — the weakest point

The V_{BE} multiplier must hold a spread that four junctions consume: Q13, the NPN bank, the PNP bank and Q14. Its multiplication factor is

$$M = 1 + \frac{R_2}{R_{23} + RV_2} = 3.01 \quad (13)$$

and M **cannot be chosen freely**. The spread required is fixed at about 2.05 V by the target quiescent, and Q12's own V_{BE} is about 0.68 V at its operating current, so M is pinned near 3 whatever values R23 and RV2 take. Re-trimming RV2 at R23 values from 2.2 k Ω down to 470 Ω gives $M = 3.01$ in every case. The only way to raise M is to lower Q12's own V_{BE} , which means running it at a far lower current, which the divider cannot support.

$M = 3$ compensating four junctions is under-compensated; $M = 3$ compensating the two *output* junctions is over-compensated. Which case applies depends on what actually gets hot:

Scenario	I_q /bank at 27 / 50 / 75 °C	
All devices hot (drivers included)	235 / 911 / 1895 mA	worst case
Output bank + Q12 hot, drivers at ambient	235 / 456 / 793 mA	realistic
Output bank hot, Q12 not tracking	235 / 456 / 747 mA	

Table 52: Quiescent current against temperature. Q12 sits in the heatsink row between Q10 and Q5, so the middle row is the applicable one.

The bias roughly doubles per 25 °C of heatsink rise. Trim to 235 mA cold and a warm board will idle at 450 mA to 800 mA, which is 13 W to 24 W — back to the dissipation the Phase H fix was meant to escape.

Do not trust these numbers quantitatively. Three of the five vendor models have $NF < 1$, which is unphysical, and their temperature parameters (XTB, XTI, EG) are curve-fitting artefacts. What the simulation establishes is the *direction and rough magnitude*, and both are corroborated by the bench: board 1V1 was observed to go from 10 mA cold to 50 mA after fifteen minutes, a factor of five, which sits between the two simulated cases.

What to do.

1. **Trim hot.** Run at the expected duty for twenty to thirty minutes, then trim to 235 mA per bank. A cold start will then sit near 50 mA to 100 mA (84 % to 94 % crossover gain), improving as it warms. That is the right way round.
2. **Measure the real tempco** — quiescent current against heatsink temperature, on the bench, with a thermocouple. It is a ten-minute measurement and it replaces a simulated number that should not be trusted.
3. **Set a re-trim threshold.** If warm idle exceeds about 400 mA (12 W), re-trim.
4. **Next revision:** tighten Q12's thermal coupling. It is already in the heatsink row, but bolting it to an output device's tab rather than merely sharing the sink would reduce the lag. Emitter degeneration in the multiplier allows the tempco to be trimmed independently of the spread, which is the standard remedy for exactly this mismatch.
5. **Note the interaction with R_1 .** Lowering R1 to 47 Ω raises driver dissipation from 0.15 W to 0.31 W each, which nudges the system toward the all-hot row. At 0.31 W on a shared sink the shift is small, but it is the reason not to go below 47 Ω .

12.5 Plant frequency response

This is what the SIM960 has to control: the small-signal voltage transfer from COMMAND to COIL_DRIVE at the zero-current operating point.

Frequency	$ H $	Phase
1 Hz	−1.57 dB	+0.1°
10 Hz	−1.56 dB	+0.7°
100 Hz	−1.08 dB	+4.1°
1 kHz	−0.29 dB	+1.5°
10 kHz	−0.12 dB	+0.1°
100 kHz	−0.10 dB	−3.0°
1 MHz	+4.34 dB	−42°
−3 dB	2.21 MHz	

Table 53: Voltage-mode plant response. Flat within 1.6 dB from DC to 100 kHz, with under 5° of phase anywhere in the SIM960’s band.

The stage contributes essentially no dynamics below 100 kHz. The +4.3 dB peak near 1 MHz is the residual parasitic resonance, an order of magnitude above the SIM960’s 100 kHz ceiling and two orders above any sensible loop crossover.

The pole the PI must compensate is the coil’s, at $R/2\pi L = 118$ Hz — not anything in the amplifier. Set the integrator time constant to $L/R = 1.35$ ms and the proportional gain for the desired crossover, exactly as if the stage were an ideal voltage source of gain 0.835 V V^{-1} .

12.6 Power and current sharing

With approximately 18 mV of V_{BE} spread modelled across each bank:

Output	Per-device current	Spread	V_{CE}	Hottest device	Stage
0 A	0.03 A to 0.07 A	72 %	15.0 V	1.0 W	7.0 W
5 A	0.82 A to 1.18 A	36 %	11.7 V	13.8 W	59 W
10 A	1.64 A to 2.33 A	34 %	8.4 V	19.6 W	84 W

Table 54: Dissipation and sharing. The hottest device carries 17 % more than its nominal share.

At 10 A the worst device sees 2.33 A at 8.4 V, which is 19.6 W. With $\theta_{JC} = 0.7^\circ \text{C W}^{-1}$ that is a 14°C junction-to-case rise, comfortably inside the 200 W rating and well clear of the second-breakdown boundary (4.0 A at 50 V) at this voltage. The limit is the heatsink, not the device.

The hottest ballast dissipates 1.19 W: **specify 2 W parts**, not the 0.88 W implied by assuming perfect sharing.

12.7 Summary

The one genuinely weak parameter is the bias tempco. Everything else is either good or bounded and understood.

Parameter	Value	Note
Gain	1.273 A V^{-1}	83.5 % of ideal
Zero current at	+1.08 V command	absorbed by the PID
Output at $\pm 10 \text{ V}$ command	$-11.95 \text{ A} / +11.18 \text{ A}$	19 % / 12 % margin
Nonlinearity	7.0 % of full scale	inside the loop
Crossover gain	98 %	at the recommended trim
Quiescent, cold-trimmed	235 mA/bank, 7.0 W	doubles per 25°C
Trim range	0 mA to 1036 mA/bank	useful span is 5 of 25 turns
Trim sensitivity	45 % per turn	at the target
Plant, DC to 100 kHz	flat 1.6 dB, under 5°	PI compensates the coil, not the stage
Dissipation at 10 A	84 W stage, 19.6 W worst device	heatsink-limited
Current sharing at 10 A	34 % spread	worst device 2.33 A
Disturbance sensitivity	0.0133	$6.2\times$ better than as-built

Table 55: The local optimum, characterised.

13 Architecture Assessment

With the board working, the question of whether to keep it or replace it with a published design was reopened. This section records the comparison and the decision.

13.1 Prior art in the application domain

Precision bipolar coil current drivers in atomic physics fall into a small number of recognisable families.

Single monolithic power amplifier

Sabulsky, *Rev. Sci. Instrum.* **95**, 081401 (2024), arXiv:2406.19944. A tutorial aimed explicitly at graduate students, using a single OPA549 ($\pm 8 \text{ A}$, 900 kHz gain-bandwidth, approximately $1 \text{ pA}/\sqrt{\text{Hz}}$) as a bipolar current source for bias and shim coils, non-inverting, with a current-limit resistor and direction LEDs, giving approximately $\pm 4 \text{ V}$ at 8 A directly from the chip. Derived from the Sewell and Zou theses at Imperial. This is the field’s default answer.

Push-pull monolithic

arXiv:2111.07877. Two OPA549s in push-pull at 2 A V^{-1} , current sensed on a series resistor, benchmarked against grey-molasses field compensation. Descended from the Arango/Kuang source at MIT.

Zero-drift PI plus monolithic output

Yang et al., *Rev. Sci. Instrum.* **90**, 014701 (2019). Precise series sense resistor, instrumentation-amplifier sensing, PI control on a zero-drift operational amplifier. The output stage sources or sinks with a compliance voltage greater than $\pm 12 \text{ V}$ up to 16 A, broadband, with a sub-1 Hz $1/f$ corner. An open replica exists from IQO Hannover (Martin Quensen, 2021), with KiCad sources.

Complementary MOSFET banks

NIST/JQI $\pm 20 \text{ A}$ source, *AIP Advances* **13**, 065309 (2023). Complementary N- and P-channel MOSFET banks driven by zero-drift operational amplifiers, 90 kHz bandwidth, 140 μA of noise at $\pm 20 \text{ A}$. Class-AB with trimmed gate-offset summing amplifiers to avoid crossover distortion. This is direct confirmation both that the crossover problem is real and of how the field solves it.

Large water-cooled MOSFET banks

ETH/Munich ppm Feshbach controller, arXiv:2303.13682: 450 A water-cooled bank, twenty

balanced FETs per bank with $0.05\ \Omega$ source shunts, OPA277 local loops, approximately 300 Hz bandwidth, battery power for low noise, coil snubbers. JILA BCS-BEC crossover thesis, cond-mat/0601054: water-cooled paralleled MOSFETs as servo elements plus FET switches for fast turn-off, Hall-sensor feedback.

H-bridge polarity switching

Liu et al., *Rev. Sci. Instrum.* **94**, 053201 (2023), $4 \times 10^{-9}/\sqrt{\text{Hz}}$, sensor placed close to the coil to remove the coil's parasitic LC from the loop. Also dual-species Li/Sr (arXiv:1904.05230), ^{39}K BEC with an IXYS $13\ \text{m}\Omega$ bridge (arXiv:2201.04544), and a compact Feshbach coil with a MOSFET plus RC snubber for 30 ns turn-off (arXiv:2103.05273).

On-demand high voltage

Uthoff-Rodríguez et al., *Rev. Sci. Instrum.* **96**, 104708 (2025), arXiv:2503.22554. Surveys the field and notes that monolithic power-amplifier solutions are easy to implement but restrict customisability. Benchmarks against Kepco BOP-L, Matsusada DOS and AE Techron 7200 as the commercial standards for inductive loads.

Grounding practice

Heidelberg bachelor thesis (Hönen, 2022), OPA549 plus instrumentation amplifier, with a direct discussion of SGND/PGND separation and reference-pin ground-loop pitfalls.

13.2 Comparison against the present design

	Present BJT board	Yang 2019 / OPA549	NIST complementary MOSFET
Output element	5 pairs MJL21193/94	1–2 OPA549	N/P-MOSFET banks
Current capability	$\pm 10\ \text{A}$ design, $-8\ \text{A}$ limited by R24	$\pm 8\ \text{A}$ per device	$\pm 20\ \text{A}$ demonstrated
Current limit	none	built in, pot-programmable	external
Thermal shutdown	none	built in, with status flag	external
Crossover handling	V_{BE} multiplier, trimmed	inside the device	trimmed gate-offset summing amplifiers
Sense element	DP50IP fluxgate	$10\ \text{m}\Omega$ four-terminal shunt plus AD8429	shunt
Loop	external SIM960	on-board ADA4522 PI	on-board zero-drift
Bandwidth	rail-limited; monitor flat to 1 kHz	approximately 2 kHz crossover by design	90 kHz
Parts cost	moderate, all in hand	two OPA549 plus shunt	many FETs, more complex
Documentation	this document	published paper plus open replica	published paper only

Table 56: Architectural comparison.

13.2.1 What the monolithic route buys

Protection, primarily. The OPA549 provides programmable current limit, thermal shutdown and a status flag in the package. On the present board a single device failure, a momentary short, or a command transient during power-up puts the full supply current through ten paralleled power transistors with nothing but the supply's own current limit in the way. That is the single largest reliability difference, and it is not a small one for an instrument that will be power-cycled routinely by people who did not build it.

It also buys a documented reference design. The Hannover replica has KiCad sources, and Yang's paper has measured noise figures to compare against.

13.2.2 What the MOSFET route buys

Higher current per device and no base current, so no h_{FE} dependence and no bias-network I_b term of the kind that caused fault 4. Against that, MOSFET class-AB stages have a larger and more temperature-dependent crossover region, which is why NIST uses trimmed gate-offset summing amplifiers rather than a single bias spreader. That is more circuitry, not less.

13.2.3 What the present design has that the alternatives do not

- **It works, and it is understood.** Every failure mode in it has been found, measured, and matched to a simulation. That is worth more than it sounds: a new board arrives with its own unknown faults, and the debugging effort recorded in Section 5 would be repeated in a different form.
- **Headroom by design.** Discrete devices rated 250 V to 400 V V_{CEO} mean the rails can be raised substantially without changing the output devices. The OPA549 is capped at ± 30 V total supply. Since the fundamental speed limit is forcing voltage, this is the only route to faster large steps that does not involve a switching stage.
- **Dissipation spread over ten packages.** At 10 A on ± 15 V the stage dissipates approximately 84 W, which is roughly 8 W per device across ten TO-264 packages on a common heatsink. Two OPA549s at the same current dissipate approximately 39 W each, which requires water cooling or a very large forced-air sink. Ten spread packages are thermally easier than two concentrated ones.
- **Instruments already integrated.** The SIM960 and the DP50IP are in the rack and working with this board.

13.2.4 Risk of a full redesign

A new board means: new layout, new BOM, new lead times on the OPA549 and a four-terminal shunt, a new bring-up with its own unknown faults, a new sense chain to characterise, and a new control loop to tune. The Hannover replica's own issue list flags unpopulated PI resistors (every value marked XXX, never determined, tested only to 1 A), a possible PI operational-amplifier footprint error, and rail capacitors that stay charged after power-off. It is a good starting point, not a finished design.

13.3 Decision

Keep the BJT boards. Bring up the remaining axes the same way, using the modification procedure in Section 7, and respin the board with the fixes in Section 14 rather than changing architecture.

The earlier decision to abandon this board in favour of an OPA549 rebuild was taken when the board did not work and the cause was unknown. That premise no longer holds. The reasoning recorded in `HANDOFF.md` remains valid as a description of the OPA549 option, including the PI derivation and the noise budget, and should be kept for the case where the protection argument eventually wins. It is not the current plan.

The one element of the OPA549 plan that should be adopted regardless is **on-board current limiting**, in whatever form. It is the only difference between the two architectures that represents a risk rather than a preference.

13.4 Supply architecture

The present arrangement uses four Delta ES 015-10 supplies per axis: one series pair for the main $\pm 15\text{ V}$ (PGND) and a second pair for the amplifier and sensor $\pm 15\text{ V}$ (SGND). Across six channels that is twenty-four supplies, which is not economic.

The analog section draws tens of milliamperes. A 10 A laboratory supply is substantially over-specified for it, and a fixed regulated rail is quieter than a programmable one.

Recommended arrangement: keep one ES 015-10 series pair per axis for the power stage, and give each axis its own small regulated $\pm 15\text{ V}$ for the amplifier and transducer, generated locally from a transformer secondary with 7815/7915-class regulators or an equivalent low-noise module. This reduces the laboratory-supply count from four per axis to two, and has three further benefits:

1. It resolves audit finding 2 structurally. A fixed $\pm 15\text{ V}$ regulator cannot be turned up past the DP50IP's $\pm 15.75\text{ V}$ limit, whereas a programmable supply can be, and eventually will be.
2. It preserves the per-axis single-point ground. Each axis keeps its own SGND to PGND bond.
3. Regulated rails drift and ripple less than programmable supplies.

What not to do: share one signal-side $\pm 15\text{ V}$ rail across all axes. A shared SGND combined with per-axis SGND-to-PGND bonds creates one ground loop per axis, which is precisely the failure mode the legacy prototype suffered from and which finding 3 of the audit describes. If a shared rail is unavoidable for cost reasons, then only one axis may carry the SGND to PGND bond, and the others must reference through it, which couples the axes and is worse.

A per-axis regulated rail also allows the power-stage rails to be raised later, independently and one axis at a time, without any risk to the transducers.

13.5 Sense element: fluxgate against shunt

The DP50IP is a fluxgate transducer and therefore responds to ambient magnetic fields. In an apparatus that deliberately generates magnetic fields, and where the transducer sits near the coils it is measuring, this is a legitimate concern rather than a theoretical one.

Against that, the DP50IP's specified noise is approximately $50\text{ }\mu\text{A RMS}$ over 0 kHz to 100 kHz, roughly three orders of magnitude below the 42 mA pk-pk observed in Section 8. The observation is far more likely to be a measurement artefact than a transducer limitation, and the shorted-cable test should be done before any hardware change.

If a shunt is eventually wanted, the published designs use a $10\text{ m}\Omega$ four-terminal foil shunt with an AD8429 instrumentation amplifier at $G = 101$, giving 1.01 V A^{-1} , with the shunt mounted on the cold plate because its temperature coefficient (approximately $5\text{ ppm}/^\circ\text{C}$) is a direct DC current error. A shunt is quieter at high frequency; the fluxgate is more accurate at DC, with 10 ppm linearity and drift below 2 ppm per year. The published compromise is to close the fast loop on the shunt and use the fluxgate as an independent monitor, optionally with a very slow outer loop correcting the inner setpoint.

13.5.1 Characterising the ambient field

To determine whether ambient field is actually the noise source, an oscilloscope-compatible magnetometer is needed. Three classes are relevant:

Fluxgate magnetometer probes with analog output

Stefan Mayer Instruments FL1-100 (single axis, $\pm 100\text{ }\mu\text{T}$, analog voltage output, approximately 1 kHz bandwidth) and FLC3-70 (three axis). Bartington Mag-13 or Mag-03 series with analog

output and a Spectramag interface. These measure absolute DC and low-frequency fields, which is what matters for a fluxgate current transducer, and they plug into an oscilloscope directly. This is the correct instrument for the question.

Near-field H-field probes

Beehive Electronics 100C or 101C series, into a 50Ω oscilloscope input. These are inductive loop probes: cheap, immediately available, and useful for locating AC sources such as switching supplies and mains harmonics, but they are AC-only and uncalibrated for absolute field.

Improvised pickup coil

A small multi-turn air coil into the oscilloscope, calibrated against a known Helmholtz field. Adequate for identifying whether a spectral line exists and where it is strongest, which may be all that is required.

The practical sequence is: first do the shorted-cable test, because it is free and it settles whether the noise is even magnetic. Only if the noise is confirmed to follow the transducer is a magnetometer worth buying, and then a fluxgate probe with analog output is the right choice.

14 Requirements for the Next Board Revision

Every item below is traceable to a finding in Section 4 or to a fault in Section 7. Items marked *bodged* exist as a bench modification on board 1V1 and must become a designed-in feature.

14.1 Must fix

1. **E/D bias circuit.** Permanent divider or current source holding E/D in the enable window, plus a 30 pF capacitor from E/D to a low-impedance node. *Bodged as 10 k Ω /4.7 k Ω .* (Fault 1, finding 4.)
2. **Assign a footprint to NT1,** or fit a zero-ohm link, so SGND and PGND are bonded in copper at one point. *Bodged as an external wire.* (Fault 2, finding 3.)
3. **Base stoppers on Q13 and Q14.** 220 Ω , placed so they carry only base current, with the shortest possible connection to the base pad. *Bodged via trace cuts.* (Fault 3.)
4. **Rescale the V_{BE} multiplier divider** by approximately ten: $R_2 = 1\text{ k}\Omega$, $R_{23} = 220\text{ }\Omega$, $RV_2 = 500\text{ }\Omega$. This fixes both the h_{FE} sensitivity and the $R_2 I_b$ floor in one change, and makes the trim reproducible. *Bodged as 15.6 k Ω across R_2 .* (Fault 4, finding 6.)
5. **Separate the DP50IP supply from the amplifier rails.** Give U1 its own regulated $\pm 15\text{ V}$; never share the net. Without this, raising the rails destroys the transducer. (Finding 2.)
6. **Isolation resistor between OPA455 OUT and C22,** 10 Ω to 47 Ω , or relocate C22 so it does not load the amplifier output. (Finding 5.)
7. **On-board current limiting.** VI-limiter transistors across the drivers, or an equivalent. This is the one item where the monolithic alternatives are unambiguously better, and it should not be left to the supply's current limit. (Finding 13.)
8. **Replace R24 with an active current source,** or at minimum lower it to approximately 667 Ω in a part rated at least 0.5 W. As fitted it caps negative full scale near -8 A . (Finding 18.)
9. **DC return on COMMAND,** 10 k Ω to 100 k Ω to SGND. Required even though the command is set externally: it defines the node when the source is disconnected. (Finding 8.)

14.2 Should fix

10. **Status Flag pull-up** (10 k Ω to 5 V referenced to E/D Com) plus a test point, so thermal and overcurrent protection can be distinguished from a disabled output. (Finding 9.)
11. **Gain trim on the amplifier stage.** Currently fixed at unity. A trimmable gain allows the command scaling to be matched to the transducer scale factor per axis without adjusting the SIM960.
12. **Zobel network at COIL_DRIVE,** 10 Ω in series with 100 nF to PGND. Adds high-frequency damping at the output and would further reduce the residual 3.7 MHz ring. (Finding 11.)
13. **DC bleed on COIL_DRIVE,** 10 k Ω to PGND, so the output is defined during unloaded bring-up. (Finding 12.)
14. **Commit a part number for D1 and D2** and verify the repetitive surge rating against 44 mJ per shutdown. (Finding 10.)
15. **Re-evaluate R25,** currently 100 Ω , which is exactly the DP50IP's maximum allowed burden with no margin. (Finding 14.)

16. **Route the V_{BE} multiplier with short paired traces.** Keep Q12 on the heatsink for thermal tracking but eliminate the 90 mm to 140 mm trace runs in the oscillating loop. (Finding 7.)
17. **Specify R24 at 2 W** if the rails will be raised to ± 30 V. (Finding 17.)
18. **Verify the MJL21194 footprint pin map**, which is assigned from the MJL21193G library. Almost certainly harmless, but it is a cross-reference. (Finding 15.)
19. **Test points** on HV_DRIVE_NODE, both drive buses, PNP_PREDRIVE and COIL_DRIVE. Every one of these had to be probed with a hand-held probe on a component lead during debugging.

14.3 Bring-up procedure for the remaining axes

Board 1V1 is the proven configuration. The remaining boards should be brought up in this order, which follows the sequence that worked rather than the sequence originally planned.

1. Fit the E/D divider ($10\text{ k}\Omega$ / $4.7\text{ k}\Omega$) before any power is applied. Verify approximately 4.8 V at pin 8 relative to E/D Com with signal rails only.
2. Bridge SGND to PGND at the supply, one point only.
3. Fit the $220\text{ }\Omega$ base stoppers at Q13 and Q14 via trace cuts, per Section 7. Do this *before* first applying main power, not after; the board will otherwise oscillate and sit at current limit, which is what consumed the original debug.
4. Fit $15.6\text{ k}\Omega$ across R2.
5. Set RV2 to maximum resistance, which is minimum bias, as the starting point.
6. Confirm the amplifier is a working follower with signal rails only: command 4 V should give 4 V at HV_DRIVE_NODE.
7. Apply main rails at a low current limit with no load. Scope HV_DRIVE_NODE. It must be clean; any oscillation means a stopper is not carrying only base current.
8. Trim RV2 for 10 mA of supply current cold, then let it warm for fifteen minutes and re-trim if it exceeds approximately 100 mA.
9. Fit a resistive test load. Confirm linearity and phase against command from 10 Hz to 1 kHz.
10. Verify D1, D2 and D3 are fitted, then connect the coil.
11. Sign check: a positive command must produce a positive DP50IP reading before the SIM960 loop is closed. Closing a loop with the wrong sign is a runaway.

14.4 Open measurements

- Two-point DC sweep to separate slope from offset and explain the 2.1 V measured offset against the 1.2 V nominal.
- Shorted-cable noise test to determine whether the 42 mA pk-pk noise floor is the transducer or the measurement.
- FFT of the monitor channel with a Hanning window, to test whether 79.4 Hz is a real line.
- Noise against operating point: DC commands of -4 , 0 and $+4$ V with monitor AC-RMS recorded at each, to test whether noise tracks polarity.
- Slow the command edge to approximately $10\text{ }\mu\text{s}$ and re-run test 7. If the 3.7 MHz ring disappears, it was pickup rather than real current.

- Operation above 1 A. Everything measured so far is under 100 mA, which leaves the output stage and the transducer far below their useful range and makes the signal-to-noise ratio far worse than it needs to be. A load of a few ohms, or several primary turns through the transducer, would fix this.
- Current sharing: voltage across each of the ten $0.22\,\Omega$ ballasts at approximately 2 A per bank. Expect approximately 88 mV each. A large spread indicates V_{BE} mismatch and is worth knowing before running 10 A.
- Negative full scale, to test the predicted -8 A R24 limit.
- Rails to full $\pm 30\text{ V}$ — only after the DP50IP supply has been separated.

15 Retracted Conclusions and Debugging Methodology

15.1 Retracted conclusions

Every entry below was stated as a conclusion at some point and later shown to be wrong. They are recorded so they are not repeated, and because the pattern in them is more useful than any individual entry.

Claim	Why it was wrong
The original 5 A idle was a ground loop	It was the disabled amplifier. With a shared supply the grounds <i>were</i> tied through the supply common, so the collapsed output dragged the predrive network to the negative rail. Unplugging any lead broke that path, which mimics ground-loop behaviour
D5 and D6 are wired backwards	Read from a low-resolution CAD screenshot. Cathode bands and the netlist both show all five correct
Change R23 to fix the bias range	The trim range is adequate; the target lands at approximately 25 % of RV2 travel. The problem was the floor, not the range
Move the $-IN$ tap to COIL_DRIVE	The current loop is closed by the SIM960; the offset is absorbed by design. The suggestion would have put the output stage inside a loop that does not need it, and required compensation
Add series resistance at the trimmer to reach a deadband	Raising resistance at RV2 increases $R_2 I_b$ as fast as it reduces MV_{BE} . Simulation showed even 47 k Ω bottoms out at 16 mA per bank
RV2 has no authority	Junction voltage is logarithmic: 10 mV is a factor of $e^{10/26} \approx 1.5$ in current. RV2 was working; the observable was wrong
Trim bias by watching ballast millivolts	Approximately 10 mV across 0.22 Ω is at the multimeter's floor, and the readings over-counted by 70 % against the supply meter. Use the supply current meters
27.8 / 38 MHz driver- f_T oscillation	Undersampled captures. The real modes are 1.67 and 3.09 MHz
36 ns square wave at HV_DRIVE_NODE	Same cause: the first capture was undersampled and was discarded
47 Ω is the stopper sweet spot	Simulation shows poor margin
100 Ω stoppers are sufficient	True only near zero current. With vendor models the PNP half oscillates at -1 V and -3 V commands. 200 Ω minimum; 220 Ω fitted
Q12 is not thermally coupled	The PCB shows Q12 in the heatsink row between Q10 and Q5
The long Q12 traces cause the PNP oscillation	Shortening them in simulation did not help. The PNP devices' own f_T asymmetry does
C22, with or without a series resistor, helps stability	It does not; in every variant it reduced margin. With 220 Ω stoppers it is harmless and stays fitted
The design has no SGND to PGND tie	It does, via NT1. NT1 simply has no footprint
The gain is 6.5 times low	Unit error: 0.4 A V $^{-1}$ used where the scale is 0.4 V A $^{-1}$, understating every current by 6.25 times. The gain is within 2 % of the load prediction
Adding <code>gminsteps</code> and <code>srcsteps</code> will help convergence	It broke cases that had previously converged

Table 57: Retracted conclusions.

15.2 Methodology notes

1. **Verify polarity from the netlist or the physical part, never from a CAD screenshot.** A pixel-level read of a rendered symbol is not evidence. The retracted D5/D6 finding cost hours and sent the investigation toward components that were correct.
2. **Check internal consistency before building a theory.** Several dead ends came from numbers that did not add up: a 0.12 V V_{BE} alongside a claimed 29 mA of conduction; 295 mA

inferred through the ballasts while the supply delivered 170 mA. An inconsistency is information; it says one of the measurements is being misread.

3. **A collapsed rail invalidates everything downstream.** Predrive voltages measured with the negative rail at -1.4 V were consequences, not causes. Restore the rail, or accept that the readings only describe the fault condition.
4. **Constant voltage across a wide current range means junction clamping; voltage proportional to current means a resistive short.** 1.4 V at 0.17 A against 1.2 V at 4 A is what ruled out a hard short and redirected the search to active devices.
5. **Equal current on both supplies means shoot-through, not load current.** Load current appears on one rail and returns through ground. This single observation located the conduction in the output banks.
6. **Know which observables are linear and which are logarithmic.** Trimming bias by watching a junction voltage hides a 1.5-fold current change in 10 mV . Ballast resistors are linear in current; junction voltages are not.
7. **Do not skip "is it even enabled?" on parts with shutdown pins.** The datasheet said a floating E/D self-enables. A $1\text{ }\mu\text{A}$ pull-up against a 0.8 V threshold does not survive real-world leakage, and the datasheet says so in the same section.
8. **Distinguish "disabled" from "dead" from "protected" by the signature, not by guessing.** A dead output gives a rail. A disabled output stage gives an attenuated but tracking voltage, because the output impedance rises while the inputs stay live. Overtemperature and overcurrent give the same high impedance as shutdown, which is why finding 9 matters.
9. **Get the netlist early.** One parse answered questions that days of in-circuit probing could not, and produced three latent bugs that had nothing to do with the fault being chased. In-circuit resistance and diode measurements on a densely paralleled stage are ambiguous by construction, because every measurement sees parallel paths.
10. **A second independently assembled board is a diagnostic instrument.** Identical symptoms on two boards with separately verified components ruled out component damage in one step and redirected the entire investigation to the design.
11. **Test-setup conditions are part of the circuit.** COIL_DRIVE was shorted to PGND through the sense path for much of this debug, which is the worst-case load for a class-AB stage and materially changed the symptoms. Recording the load with each measurement is not bookkeeping; the load is a circuit element.
12. **A handheld multimeter in AC mode will not find a megahertz oscillation.** Its AC bandwidth stops far below. The DC reading wandering between 0.1 V and 1 V was the only hint available before the oscilloscope was used, and it is a weak one.
13. **Check the sampling rate before believing a frequency.** Two separate oscillation frequencies were misreported from undersampled captures. If the waveform looks like a clean square wave at the very edge of the timebase, it is aliased.
14. **Check units before concluding that performance is wrong by a large factor.** A gain reported as 6.5 times low was a reciprocal error in the sense scale. A discrepancy that happens to be near the square of a scale factor is a units error until proven otherwise.
15. **Simulate the fix before fitting it, and simulate across the full operating range.** The $100\text{ }\Omega$ stopper value was stable at zero command and would have appeared to work on the bench at idle, then oscillated the first time the command went to -3 V . Sweeping the command was what caught it.

16. **Validate the model against something already measured before trusting its predictions.** Reproducing the oscillation frequency and amplitude to within 4 % and 3 %, with no parameter fitted to that data, is what justified acting on the stopper and bias predictions.

16 Reference Calculations

16.1 Load

$$L = \tau R_{dc} = 1.35 \text{ ms} \times 0.656 \Omega = 0.886 \text{ mH} \quad (14)$$

$$f_{\text{pole}} = \frac{R_{dc}}{2\pi L} = 118 \text{ Hz} \quad (15)$$

$$E = \frac{1}{2} L I^2 = \frac{1}{2} \times 0.886 \text{ mH} \times (10 \text{ A})^2 = 44 \text{ mJ} \quad \text{per shutdown} \quad (16)$$

$$V_{\text{compliance}} = I R_{dc} = 10 \text{ A} \times 0.656 \Omega = 6.56 \text{ V} \quad (17)$$

Forced discharge through D1 or D2 into the 1000 μF rail capacitors gives $dI/dt \approx 34\,700 \text{ A s}^{-1}$, which is approximately 290 μs to zero from 10 A.

16.2 Bias network

$$M = 1 + \frac{R_2}{R_{23} + R V_2} = 1 + \frac{10 \text{ k}\Omega}{2.2 \text{ k}\Omega + 0 \text{ k}\Omega \text{ to } 5 \text{ k}\Omega} = 2.39 \dots 5.55 \quad (18)$$

$$V_{\text{spread}} = M V_{BE}(\text{Q12}) + R_2 I_b(\text{Q12}) \quad (19)$$

The trimmable term spans 1.55 V to 3.6 V against a target of approximately 2.4 V to 2.6 V for four junctions, landing at $R V_2 \approx 1.1 \text{ k}\Omega$ to $1.5 \text{ k}\Omega$, about 25 % of travel. Maximum pot resistance gives minimum spread and is the safe starting point.

The floor term with $R_2 = 10 \text{ k}\Omega$ and $I_b = 0.05 \text{ mA}$ to 0.28 mA contributes 0.5 V to 2.8 V, in practice 0.5 V to 0.8 V. With $R_{2,\text{eff}} = 5.86 \text{ k}\Omega$ this term is reduced by 41 % and M falls with it.

Divider current, approximately 0.18 mA, against Q12 base current 0.05 mA to 0.28 mA: a ratio below two, where the design rule asks for at least ten.

16.3 Speed limit

$$\Delta t = \frac{L \Delta i}{V_{\text{forcing}}} \implies 10 \text{ A in } 591 \mu\text{s on } \pm 15 \text{ V} \quad (20)$$

The full rail table is in Section 1.3. The DC compliance requirement of 6.56 V is easily met on $\pm 15 \text{ V}$; the limitation is purely dynamic.

16.4 Sense chain

$$\text{DP50IP, 4 turns} \Rightarrow 1:250, 12.5 \text{ A nominal} \quad (21)$$

$$K_s = \frac{R_{25}}{250} = \frac{100 \Omega}{250} = 0.4 \text{ V A}^{-1} = 2.5 \text{ A V}^{-1} \quad (22)$$

$$10 \text{ A} \Rightarrow 40 \text{ mA secondary} \Rightarrow 4.0 \text{ V}; \quad 12.5 \text{ A} \Rightarrow 5.0 \text{ V} \quad (23)$$

16.5 Miscellaneous

- E/D divider: $15 \text{ V} \times 4.7 \text{ k}\Omega / (10 \text{ k}\Omega + 4.7 \text{ k}\Omega) \approx 4.8 \text{ V}$; approximately 1.0 mA of pull-up against a pin draw of approximately 50 μA .
- Ballast readout: $V/0.22 \Omega = \text{device current}$. 13 mV is 59 mA; 24 mV is 109 mA; 88 mV is 400 mA. At 2 A per bank each device carries 400 mA and drops 88 mV.

- Junction voltages are logarithmic: $\Delta V_{BE} = 10 \text{ mV}$ corresponds to $e^{10/26} \approx 1.5$ times the current at room temperature.
- Output offset as built: $V_{\text{COIL_DRIVE}} \approx V_{\text{COMMAND}} - 1.2 \text{ V}$ nominal, 2.1 V measured. Into 0.656Ω the nominal figure is approximately 1.8 A of error at zero command in open loop, zero in closed loop.
- Triangle-wave variance: $\sigma^2 = V_{pp}^2/12$. Used to extract signal amplitude from a noise-dominated capture.
- Stage dissipation at 10 A on $\pm 15 \text{ V}$: approximately 84 W , or 8.4 W per output device across ten packages.

A Modification Register, Board 1V1

#	Change	Status	Reversible
1	E/D divider: 10 k Ω from +15V.OP to U2 pin 8; 4.7 k Ω pin 8 to SGND	Keep; permanent fix required on respin	Yes
2	SGND to PGND bridged at the supply, one point	Keep; NT1 footprint required on respin	Yes
3	D2 lead cut during debugging	Restored	—
4	Load: ten 1 k Ω in parallel, later 113.2 Ω non-inductive	Test only	—
5	C22 lifted during debugging	Refitted ; stable with 220 Ω stoppers	—
6	220 Ω base stoppers on Q13 and Q14 via trace cuts	Keep; designed-in on respin	Yes
7	15.6 k Ω across R2	Keep; divider rescale on respin	Yes
8	D3 refitted	Done	—
—	RV2 trimmed for 50 mA warm	Re-check hot after any change	—
—	Rails at ± 15 V throughout	Planned permanent	—

Table 58: Physical modifications to board 1V1, current state.

B SPICE Deck Inventory

File	Contents
spice_deck_ generator.py	Parameterised deck builder. Switches: c22, c22val, c22r, rstop, rstop14, rv2, r2p, r24, d2, zobel, load (bank / coil / coilfast / short), parasitics, pscale, layout (Q12 trace inductances), real (vendor models), opa (macro / ti), bf12 (Q12 h_{FE}), mode (tran / dc / op), ikick, kick, kick_t, cmd.dc, cmd.wave, cmd.ramp, nodeset
spice_as_built.cir	As-fabricated configuration: C22 fitted, no stoppers. Oscillates
spice_recommended_ fix.cir	C22 lifted, 100 Ω stoppers. Superseded by the 200 Ω result but kept as the Phase F reference case
seeded.py	Operating-point extraction from the macromodel version, for .nodeset seeding of the vendor-model runs
tirun.py	Transient runner for the TI OPA455 macromodel
tiramp.py	Command-ramp runner for large-signal sweeps
spice_models/	onsemi vendor .lib models: BD139, MJE15032, MJE15033, MJL21193, MJL21194; TI opa455.lib

Table 59: Simulation file inventory.

C Source Documents

D Blank Measurement Tables

For use when bringing up the remaining axes.

File	Contents
BJT_PUSH_PULL.net	KiCad Eeschema 10.0.4 netlist, 80 components, 46 nets. The authoritative connectivity reference for this board
BJT_PUSH_PULL-davidpc.kicad_pcb	PCB layout. Source of the trace-length figures used for parasitic extraction and of the Q12 placement that retracted audit finding 7
yang2019.pdf	Yang et al., <i>Rev. Sci. Instrum.</i> 90 , 014701 (2019)
martin1.pdf, martin2.pdf	IQO Hannover open replica of Yang 2019: control board and OPA549 board
sbomb81b.zip	TI OPA455 PSpice macromodel

Table 60: Source documents.

Time from cold	+ supply	– supply	Note
0 min			
5 min			
15 min			
30 min			

Quiescent against temperature

Transfer and deadband

Edge response

Stability against operating point

Current sharing at approximately 2 A per bank

Expect approximately 88 mV on each. A large spread indicates V_{BE} mismatch and should be known before running 10 A.

Test	Load	Freq	Ampl.	Deadband	$I = 0$ cmd	Gain	Asym.
T1		10 Hz	± 1 V				
T2		10 Hz	± 4 V				
T3		100 Hz	± 4 V				
T4		1 kHz	± 2 V				

Test	Load	Step	Rise 10–90%	Overshoot	Ring	Crossover
E1		± 0.5 V				
E2		0 V to 4 V				
E3		–2 V to 2 V				
E4	coil	0 V to 4 V				

E Raw Capture Log, 25 September 2026

This is the unprocessed record of the triangle and pulse captures, kept separately from the analysed figures in Section 8 so that the raw numbers are available if the analysis is ever revisited. Board 1V1, $113.2\,\Omega$ non-inductive load, command from the function generator CH1, monitor from the DP50IP BNC. Generator CH2 supplied a $5.000\text{ V} / 4.998\text{ V}$ pulse with a very long period, acting as a DC supply for the E/D pin.

Scale used throughout: 0.4 V A^{-1} , that is 2.5 A V^{-1} . "Ideal" means $V_{\text{cmd}}/113.2\,\Omega = 8.83\text{ mA V}^{-1}$, equivalently 22.1 mV V^{-1} on the monitor channel.

Scope readings that are not signal measurements

Several automatic measurements in these captures are corrupted by the noise floor or by cursor placement and must not be used. They are listed so that nobody extracts them later.

- The $1/|dx| = 2.5\text{ kHz}$ appearing in most screenshots comes from cursor placement, not from the signal. The cursors stayed at 3.6 ms and 3.2 ms across captures.
- The monitor channel's duty, overshoot, preshoot, rise/fall and slew-rate readings are corrupted by the noise. A duty cycle of 34 % at 10 Hz and 77 % at 1 kHz is impossible for a clean triangle, which confirms the readings are noise-triggered.
- The CH2 average readings (217 mV, 171 mV, –63 mV) come from averaging over a non-integer number of periods, not from a real command offset.
- For the monitor channel, amplitude and variance are the only reliable metrics. A triangle wave has variance $V_{pp}^2/12$, which reproduces CH2's 7.78 V exactly. Applying the same relation to the monitor channel and subtracting the noise variance gives approximately 29 mV to 30 mV pk-pk at 10 Hz and 100 Hz, which is the figure used for the gain extraction in Section 8.
- The 79.4 Hz counter reading in capture 0 is almost certainly the burst envelope rather than a spectral line, since the waveform is bursty with high-frequency fill and 79.4 Hz is not a harmonic of 50 Hz or 60 Hz.

Qualitative observations not carried into the analysis

- In the triangle captures the noise appears heavier near the negative command peaks, which would point to a PNP-bank contribution. However, the $\pm 0.5\text{ V}$ square showed more noise on the positive side. These two observations contradict each other, so both are treated as hints rather than findings. The polarity-dependence test in Section 14 settles it.

Command						Current						HV_DRIVE_NODE clean						COIL_DRIVE clean					
-3 V																							
-1 V																							
0 V																							
1 V																							
3 V																							

NPN ballast						PNP ballast					
R5	R7	R11	R15	R19		R4	R8	R12	R16	R20	
mV						mV					

- A consistent offset of -3 mV to -4 mV (-7.5 mA to -10 mA) is present in every capture. Origin unresolved between a transducer zero offset and a small quiescent imbalance between banks.

#	Command (gen CH1)	Timebase	Monitor channel, as read	Ideal
0	none, CH2 flat	20 ms/div	16.8 mV pk-pk noise (cursors -11.2 / $+5.6$ mV), centre about -3 mV. Counter reads 79.4 Hz / 12.60 ms. Bursty envelope with high-frequency fill	0
1	10 Hz triangle, ± 1 V	20 ms/div, CH2 1 V/div	17.2 mV pk-pk, essentially the noise floor. A faint undulation is visible, consistent with about 7 mV pk-pk buried in the noise	17.7 mA pk-pk (44 mV pk-pk)
2	10 Hz triangle, ± 4 V	20 ms/div, CH2 5 V/div	V_{pp} 42.0 mV (noise included), AC-RMS 9.1 mV, average -4.36 mV, variance $83.5 \mu\text{V}^2$. CH2: V_{pp} 8.20 V, variance 5.05 V^2 . Clear triangle, in phase	70.7 mA pk-pk (177 mV pk-pk)
3	100 Hz triangle, ± 4 V	2 ms/div, CH2 2 V/div	V_{pp} 44.8 mV, amplitude 25.7 mV, average -2.81 mV, variance $89.3 \mu\text{V}^2$. CH2: amplitude 7.78 V, slew $\pm 1.60 \text{ kV s}^{-1}$. No visible phase lag	70.7 mA pk-pk
4	1 kHz triangle, ± 4 V	200 μs /div, CH2 2 V/div	V_{pp} 46.4 mV, amplitude 25.7 mV, average -3.79 mV, variance $124 \mu\text{V}^2$. CH2: amplitude 7.77 V, slew $\pm 15.9 \text{ kV s}^{-1}$. Lag under 20 μs , under 7° . Noise looks heavier	70.7 mA pk-pk
5	10 Hz square, ± 0.5 V	100 μs /div, CH2 2 V/div	Step of about 2 mV to 3 mV at the edge. Noise envelope grows after the edge. CH2 edge under 2 μs	8.8 mA (22 mV)
6	10 Hz square, 0 V to 4 V	1 μs /div	Step about 12.4 mV (cursors -3.2 mV to $+9.2$ mV over 1 μs). Damped ringing at roughly 4 MHz, then a second ring burst about 3.5 μs later	35.3 mA (88 mV)
7	10 Hz square, ± 2 V	500 ns/div	Step about 12 mV to 14 mV. About 30 % overshoot. Counter reads 3.70 MHz / 270 ns ring. Settles in about 1.1 μs . Three visible ring cycles. CH2 rise under 45 ns	35.3 mA (88 mV)

Table 61: Raw capture log, as read from the oscilloscope.

F Validating the Simulated Transfer Function on the Bench

The simulation programme produced a DC transfer characteristic, a crossover region, an idle current against R2 curve, a step response into the coil and a stability map against command voltage. None of these has been reproduced on the bench at a meaningful current. This appendix specifies what is needed to close that gap.

F.1 The measurement problem, and the two changes that fix it

The 25 September data cannot validate any of the simulated curves, for two reasons that have nothing to do with the board.

The load is wrong. At 113.2Ω , a 10 V command produces 70 mA. Split across five devices per bank that is 14 mA each, against a 50 mA quiescent point. The output stage never leaves its idle region, so *nothing about its large-signal behaviour is being exercised*: not linearity at current, not crossover under load, not current sharing, not the R24 starvation limit, not the thermal behaviour. The simulated curves all describe a regime the board has never been operated in.

The instrument is wrong. Extracting a gain from a triangle buried in 17 mV pk-pk of noise required a variance argument to recover 29 mV of signal. A DC measurement with a multimeter averages that noise away entirely: a $6\frac{1}{2}$ -digit meter on DC volts integrates over many power-line cycles and will read the monitor to microvolts. **Every linearity and transfer point should be taken as a DC measurement with a meter, not extracted from an oscilloscope trace.** The oscilloscope is for edges, ringing and oscillation; the meter is for accuracy.

F.2 Building a load that exercises the stage

The target is a load that draws amperes at a few volts and survives it. Three options, in order of preference.

The real coil.

$0.656\ \Omega$, so 10 A needs 6.56 V and dissipates 43 W in the coil itself. This is the actual load and its DC resistance is known, which makes it the best choice for the DC transfer measurement — with the caveat that DC points must be short dwells with airflow, since the coil is designed for pulsed operation, not sustained 10 A DC. Start at 1 A and work up.

A chassis-mount power resistor.

A $1\ \Omega$ or $0.5\ \Omega$ aluminium-housed wirewound rated 100 W, bolted to any scrap heatsink, costs a few dollars and is the right tool. $1\ \Omega$ gives 5 A at 5 V (25 W), which exercises the stage properly without approaching full scale. Non-inductive types exist and are preferable, but ordinary wirewound is acceptable for DC points.

An array of quarter-watt resistors.

Possible but impractical. To reach $2\ \Omega$ at 2 A (8 W total) with sensible derating needs of order a hundred parts in parallel, and the resulting splay inductance and uneven current sharing make it a poor load. The bench bank used for debugging (ten $1\ \text{k}\Omega$ in parallel) was appropriate for *finding* the fault and is not appropriate for characterising the fix.

F.3 Measurement 1 — DC transfer and linearity

This is the measurement that resolves the outstanding 2.1 V offset question and validates the simulated transfer curve in one sweep.

1. Command from the SIM960 in manual mode or a DC source, stepped through $-10, -8, -6, -4, -3, -2, -1, -0.5, 0, +0.5, +1, +2, +3, +4, +6, +8, +10\ \text{V}$.
2. At each point record, with a meter: the command voltage at J1; the monitor voltage at J13; the voltage directly across the load; and both supply currents.
3. Compute the current two ways — from the monitor at $0.4\ \text{V A}^{-1}$ and from the load voltage divided by the measured load resistance. The two should agree. If they do not, the transducer scale factor is not what it is assumed to be, which is precisely the ambiguity that produced the retracted 6.5-times-low conclusion.
4. Fit a straight line over the linear region and record slope, intercept and the residual from the fit.

What this delivers: the true gain in A V^{-1} , the true offset separated from the slope, the integral nonlinearity, and a direct check on the transducer scale. Expected: slope within a few percent of unity referred to the load, intercept near $-1.2\ \text{V}$ to $-2.1\ \text{V}$ of command.

F.4 Measurement 2 — crossover region

The simulated zero-crossing detail cannot be seen with 0.5 V command steps.

1. With the command offset so the output sits at zero current, step in 25 mV increments over a $\pm 0.5\ \text{V}$ window around that point.
2. Record monitor voltage at each step with the meter.

3. Differentiate numerically to get incremental gain against operating point. The crossover signature is a dip in that derivative, not a flat spot in the output.

Simulation predicts the dip depth depends strongly on quiescent current: approximately 5% at 300 mA per bank rising to 90% at 2 mA per bank. At the present 50 mA setting the dip should be clearly visible. **Repeat the whole sweep at two or three RV2 settings** — this maps quiescent current against crossover linearity directly, which is the trade-off the bias trim exists to make, and it is the most useful single measurement remaining on this board.

F.5 Measurement 3 — idle current against R2

Partially done. To complete the comparison with the simulated curve, record supply current against RV2 position (in turns from one end) for the present $R_{2,\text{eff}} = 5.86 \text{ k}\Omega$, hot and cold. If a second board is available with an unmodified $10 \text{ k}\Omega$ R2, record the same curve there. Two curves are enough to confirm the $R_2 I_b$ floor term of Equation 5 quantitatively rather than qualitatively.

F.6 Measurement 4 — step response into the real coil

With the coil connected and the SIM960 still in manual mode, apply a command step and record the monitor on the oscilloscope at 1 ms/div.

Expected: a single exponential with $\tau = L/R = 1.35 \text{ ms}$, reaching 63% in that time, with no overshoot and no oscillation. Any deviation from a clean first-order exponential means the output stage is contributing dynamics of its own, which is what the step-response simulation was built to check.

Also repeat the $\pm 2 \text{ V}$ fast-edge test from capture 7 **with the coil connected**, to settle the question in Section 8.4: whether the 3.7 MHz ring survives into the real load, and at what amplitude.

F.7 Measurement 5 — stability map against command

This validates the stopper sweep, which is the simulation result the whole fix rests on and which has been confirmed only at idle.

Set a DC command at each of $-10, -6, -3, -1, 0, +1.5, +3, +6, +10 \text{ V}$ into a load that draws real current, and at each point scope HV_DRIVE_NODE and COIL_DRIVE at 100 ns/div. Record clean or not clean. Simulation says 220Ω is clean at every one of these points and that 100Ω would fail at -1 V and -3 V ; the negative points are therefore the important ones and should be taken first.

F.8 Measurement 6 — negative full scale and R24 starvation

Sweep the command negative in 1 V steps while recording the voltage from HV_DRIVE_NODE to PNP_PREDRIVE (the predrive spread). Simulation predicts this collapses from approximately 2.2 V to 0.9 V as Q12 saturates, capping negative output near -8 A to -9.5 A . Finding the knee confirms audit finding 18 and sets the value R24 must be changed to.

F.9 Measurement 7 — current sharing

At approximately 2 A per bank, measure the voltage across each of the ten 0.22Ω ballasts. Expect approximately 88 mV each. Use a meter, not the oscilloscope — the earlier attempt to trim bias on ballast millivolts failed because 10 mV across 0.22Ω is at the handheld meter's floor, but 88 mV is comfortably above it.

A spread greater than approximately $\pm 15\%$ indicates V_{BE} mismatch severe enough that one device carries disproportionate current, which is the failure mode that destroys paralleled banks. This must be known before running 10 A.

F.10 Measurement 8 — noise, properly

Independent of the transfer function, and the highest-value measurement remaining, because the noise floor is in the band the coil does not attenuate.

1. Capture the monitor channel under three conditions: board off with the transducer powered; the monitor BNC shorted at the oscilloscope; everything on at zero command.
2. Use the oscilloscope's FFT with a Hanning window rather than the frequency counter, to determine whether 79.4 Hz is a real spectral line or a burst envelope.
3. Record monitor AC-RMS at DC commands of -4 , 0 and $+4$ V. If the noise tracks polarity, it originates in the output banks; if it is constant, it is the transducer, the cable or the oscilloscope.

The shorted-cable case is the decisive one and takes under a minute. If the noise survives a shorted input it is the oscilloscope or the cable, and the transducer is exonerated — which would also make the question of replacing the fluxgate with a shunt moot.

F.11 Order of work

Measurement 8 first, because it is free and it determines whether any of the other measurements can be trusted at low signal levels. Then build the load. Then measurements 1, 2 and 5, which together validate the transfer function, the crossover and the stability map at real current. Then 4, 6 and 7, which require the coil or full-scale current and carry more risk. Measurement 3 whenever a second board is on the bench.